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Reader’s guide (what is being built, and from what)

Three kinds of objects. Throughout, we explicitly distinguish three kinds of objects.

- **State/representation objects:** spaces X, Y, Z and maps between them (e.g. $h : X \rightarrow Y$, representations r , and canonical readouts $\pi_0 : Z \rightarrow Y$ for delays).
- **Boundary probability objects:** boundary horizons (stationary path laws) $\mathbb{P} \in \mathcal{P}(\cdot^{\mathbb{Z}})$, derived edge laws $\Gamma^{(\Delta t)} := (\pi_{0, \Delta t})_{\#} \mathbb{P} \in \mathcal{P}(\cdot \times \cdot)$ for chosen lags/resolutions Δt (often over a shared horizon H), pushforwards $(\cdot)_{\#}$, boundary maps ∂_{∞} and $\partial_{\Delta t}$, compatibility fibres $\text{Fib}(\mathbb{P})$ and $\text{Fib}_{\varepsilon, H}(\mathbb{P})$, and (when it exists) Markov completion as a closure/limit.
- **Information-geometric objects:** statistical model families inside $\mathcal{P}(\cdot)$ equipped with divergence/Fisher geometry, and IG leaves (equivalence classes defined by preserved IG structure).

Boundary horizon (primitive). The only empirical primitive we assume is a *boundary horizon*: a stationary path law

$$\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}}) \quad (\text{equivalently, a Kolmogorov-consistent projective family of finite-dimensional marginals}).$$

For any chosen lag/resolution Δt (and any agreed coarse-graining, if present), we *derive* an edge record

$$\Gamma_Y^{(\Delta t)} := (\pi_{0, \Delta t})_{\#} \mathbb{P}_Y \in \mathcal{P}(Y \times Y),$$

where $\pi_{0, \Delta t}$ projects a bi-infinite path to the pair $(Y_0, Y_{\Delta t})$ (with $\Delta t \in \mathbb{N}$ in discrete time). Thus a single $\Gamma_Y \in \mathcal{P}(Y \times Y)$ is never primitive “by decree”: it is a statistic of the shared horizon. All “interior” objects are judged by how they reproduce the boundary horizon (or an agreed family of derived edge records across a shared set of lags), without assuming any completion a priori.

Realization, fibre, leaf, completion. A *realization* is an interior space equipped with a stationary path law together with a boundary readout map. The *boundary horizon map* ∂_{∞} sends a realization to its induced observed path law on $Y^{\mathbb{Z}}$, while each *derived edge map* $\partial_{\Delta t}$ extracts an edge record $\Gamma_Y^{(\Delta t)} \in \mathcal{P}(Y \times Y)$ at a chosen lag/resolution Δt . The *compatibility fibre* over the horizon, $\text{Fib}(\mathbb{P}_Y)$, is the class of realizations whose boundary horizon equals (or approximately equals, across a shared set of lags) the observed horizon; the one-lag fibre $\text{Fib}(\Gamma_Y)$ is recovered once a particular Δt^* is fixed (Definition 3). An *IG leaf* is an equivalence class in the realization space obtained by identifying realizations that induce the same information–geometric structure (information-geometric level) on their boundary models. A *completion* is *not assumed*: it is *derived* as a limit/closure when the boundary-derived operators are semigroup-consistent across resolutions.

Outline (minimal working-memory ordering)

1. §1: define the boundary horizon, derived edge records, realizations, boundary maps, compatibility fibres, and “delay is an ” (boundary *o* state/representation) with the key commuting square.
2. §1.2: introduce only the IG notions that we actually use for comparing realizations: divergence $\Rightarrow$ Fisher metric $\Rightarrow$ e/m -projections $\Rightarrow$ Pythagorean identity, and then define IG leaves (information-geometric level).

3. §1.4: formulate completion as a closure problem (approximate semigroup $\Rightarrow$ generator limit $\Rightarrow$ semigroup) rather than an assumption (boundary-probability level).
4. The remainder of the note develops the neutral-operator framework and worked examples, now referenced back to the primitives above.

Commitment ledger (what is assumed where)

state/representation level: state/representation assumptions (only when invoked)

- Measurability of maps (h, ρ, π_0) ; smoothness is *optional* and only invoked for statements about topological/coordinate fidelity.

boundary-probability level: boundary-probability assumptions (default viewpoint)

- Existence of a boundary horizon $\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}})$ (or an equivalent Kolmogorov-consistent projective family) and pushforwards of boundary statistics along measurable readouts.
- Edge laws $\Gamma_Y^{(\Delta t)} \in \mathcal{P}(Y \times Y)$ at specific lags/resolutions are *derived* from the horizon once Δt (and any coarse-graining) is fixed.
- Markov kernels/semigroups are *not assumed*; they are derived under explicit semigroup-consistency and generator-limit hypotheses across a shared horizon of resolutions.

information-geometric level: statistical/IG assumptions (only for IG comparisons)

- Dominated statistical families with smooth densities so Fisher information and α -connections exist.
- IG equivalence/leaf statements refer to preserved IG structure on *families* (not to smooth coordinate changes of state spaces unless stated).

1 Boundary-first core objects

Roadmap. We begin with the *boundary horizon* (a stationary path law, equivalently a consistent family of finite-dimensional marginals) as the only primitive empirical object; stepwise edge laws are derived once a lag/resolution is fixed. We then define *realizations* and the associated *boundary horizon/edge maps*, so that the *compatibility fibre* is simply the preimage of the observed horizon (or, after fixing a lag, of a derived edge record). Finally, we show how step operators and (when limits exist) continuous-time Markov completions arise as *derived* objects rather than assumptions. We now introduce the *minimal* objects used throughout, and we use each immediately.

1.1 Boundary horizon and boundary maps (boundary-probability level)

Definition 1 (boundary-probability level: boundary horizon). *Let Y be a measurable space. A boundary horizon is a stationary path law*

$$\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}}).$$

Equivalently, one may specify a Kolmogorov-consistent projective family of finite-dimensional marginals $\{\mathbb{P}_{t_0, \dots, t_k} \in \mathcal{P}(Y^{k+1})\}_{k \geq 0, t_0 < \dots < t_k}$. For any chosen lag/resolution Δt (an integer in discrete time), the derived edge record at lag Δt is

$$\Gamma_Y^{(\Delta t)} := (\pi_{0, \Delta t})_{\#} \mathbb{P}_Y \in \mathcal{P}(Y \times Y),$$

where $\pi_{0, \Delta t} : Y^{\mathbb{Z}} \rightarrow Y \times Y$ is the coordinate projection $\pi_{0, \Delta t}((y_t)) = (y_0, y_{\Delta t})$. When a particular lag Δt^* is fixed by the observational protocol, we suppress the superscript and write $\Gamma_Y := \Gamma_Y^{(\Delta t^*)}$.

Neutrality clause: no privileged step, partition, or edge statistic. Shared experience does not canonically determine a discretization Δt , a preferred σ -algebra/partition on Y , or a unique “edge” statistic. Accordingly, we treat the horizon $\mathbb{P}_Y$ as primitive, and view each $\Gamma_Y^{(\Delta t)} \in \mathcal{P}(Y \times Y)$ as a derived boundary statistic *relative to auxiliary choices* (lag/resolution, and any agreed coarse-graining). Dynamical structure on Y (Markov kernels, semigroups) is therefore *earned* only as a closure/limit across a shared horizon of resolutions (cf. §1.4), never assumed from a privileged one-step object.

Definition 2 (boundary-probability level: realization and boundary maps). *A realization of a boundary horizon is a triple*

$$\mathbf{R} = (S, \mathbb{P}_S, q),$$

where:

- S is an interior measurable space;
- $\mathbb{P}_S$ is a stationary path law on $S^{\mathbb{Z}}$ (no Markov assumption is made);
- $q : S \rightarrow Y$ is a measurable boundary readout map.

Let $q^{\mathbb{Z}} : S^{\mathbb{Z}} \rightarrow Y^{\mathbb{Z}}$ denote the coordinatewise map,

$$q^{\mathbb{Z}}((s_t)_{t \in \mathbb{Z}}) := (q(s_t))_{t \in \mathbb{Z}},$$

and define the induced observed path law on $Y^{\mathbb{Z}}$ by

$$\mathbb{P}_Y^{\mathbf{R}} := (q^{\mathbb{Z}})_{\#} \mathbb{P}_S \in \mathcal{P}(Y^{\mathbb{Z}}).$$

Boundary horizon map. We set

$$\partial_{\infty}(\mathbf{R}) := \mathbb{P}_Y^{\mathbf{R}}.$$

Derived edge maps. For any chosen lag/resolution $\Delta t \in \mathbb{N}$, define

$$\partial_{\Delta t}(\mathbf{R}) := (\pi_{0, \Delta t})_{\#} \mathbb{P}_Y^{\mathbf{R}} \in \mathcal{P}(Y \times Y),$$

where $\pi_{0, \Delta t}((y_t)) = (y_0, y_{\Delta t})$.

Equivalently, if we define the interior lag- Δt marginal (edge law)

$$\Gamma_S^{(\Delta t)} := (\pi_{0, \Delta t})_{\#} \mathbb{P}_S \in \mathcal{P}(S \times S),$$

then

$$\partial_{\Delta t}(\mathbf{R}) = (q \otimes q)_{\#} \Gamma_S^{(\Delta t)}.$$

When a particular lag Δt^* is fixed by the observational protocol, we write

$$\partial(\mathbf{R}) := \partial_{\Delta t^*}(\mathbf{R}), \quad \Gamma_S := \Gamma_S^{(\Delta t^*)}, \quad \Gamma_Y := \Gamma_Y^{(\Delta t^*)}.$$

When only the lag- Δt^* marginal is used, we may suppress $\mathbb{P}_S$ and write the associated edge realization as (S, Γ_S, q) .

Definition 3 (boundary-probability level: compatibility fibres). Let $\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}})$ be a boundary horizon.

Horizon fibre. The compatibility fibre over the horizon is

$$\text{Fib}(\mathbb{P}_Y) := \{R : \partial_{\infty}(R) = \mathbb{P}_Y\}.$$

Edge fibres across a shared horizon of resolutions. Fix a set of lags/resolutions $H \subseteq \mathbb{N}$ representing the shared observational horizon, and let $\Gamma_Y^{(\Delta t)} := (\pi_{0,\Delta t})_{\#} \mathbb{P}_Y$ denote the corresponding derived edge records. Given a divergence D_f on $\mathcal{P}(Y \times Y)$, we define the ε -fibre controlled on H by

$$\text{Fib}_{\varepsilon,H}(\mathbb{P}_Y) := \left\{ R : \sup_{\Delta t \in H} D_f(\partial_{\Delta t}(R), \Gamma_Y^{(\Delta t)}) \leq \varepsilon \right\}.$$

Conditional one-lag fibre. When a particular lag Δt^* is fixed and we write $\Gamma_Y := \Gamma_Y^{(\Delta t^*)}$ and $\partial := \partial_{\Delta t^*}$, we recover the one-lag fibre

$$\text{Fib}(\Gamma_Y) := \{R : \partial(R) = \Gamma_Y\},$$

and its ε -version $\text{Fib}_{\varepsilon}(\Gamma_Y) := \{R : D_f(\partial(R), \Gamma_Y) \leq \varepsilon\}.$

Remark 1 (Signpost: where information geometry enters). The boundary-first core objects are probabilistic: horizon laws, derived edge laws, and their compatibility fibres. Information geometry enters only after we fix a derived edge law (a chosen lag/resolution) and regard the compatible realizations as a statistical family. See §1.2 for the IG constructions (projections, leaves, and the neutral-operator viewpoint).

1.2 IG leaves versus compatibility fibres (information-geometric level over boundary-probability level)

Definition 4 (Boundary statistical family). A boundary statistical family for a derived edge record $\Gamma_Y \in \mathcal{P}(Y \times Y)$ (at a fixed observational lag Δt^* as in Definition 1) is a smooth statistical model $\mathcal{M}_Y \subseteq \mathcal{P}(Y \times Y)$ (typically dominated by a reference measure) with $\Gamma_Y \in \mathcal{M}_Y$. It provides the ambient information-geometric structure (Fisher metric and dual connections) used to compare realizations.

Definition 5 (information-geometric level: e - and m -projections). Let $\mathcal{M}$ be a statistical manifold with divergence $D(\cdot \| \cdot)$. Given a submanifold $\mathcal{N} \subset \mathcal{M}$ and $p \in \mathcal{M}$:

- the m -projection of p onto $\mathcal{N}$ is a point $q \in \mathcal{N}$ minimizing $D(p \| q)$ over $q \in \mathcal{N}$ (when it exists),
- the e -projection of p onto $\mathcal{N}$ is a point $q \in \mathcal{N}$ minimizing $D(q \| p)$ over $q \in \mathcal{N}$ (when it exists).

Proposition 1 (information-geometric level: Pythagorean relation (pushforward-ready form)). Let $(\mathcal{M}, D)$ be a dually flat statistical manifold with Fisher metric g . Let $p, q, r \in \mathcal{M}$ such that the ∇ -geodesic from p to q is g -orthogonal at q to the ∇^* -geodesic from q to r . Then

$$D(p \| r) = D(p \| q) + D(q \| r).$$

Remark 2 (How this interfaces with the boundary-first program). In our setting, the role of p, q, r is played by boundary laws in $\mathcal{P}(Y \times Y)$: for example, $q = \Gamma_Y$ (boundary record), while p and r can be boundary images $\partial(R) = (q \otimes q)_{\#} \Gamma_S$ from different realizations. The usefulness of Proposition 1 depends on (i) choosing a divergence D_f that yields a well-behaved IG structure, and (ii) specifying the submanifolds that encode the constraints/ansätze for realizations.

Remark 3 (Preview: IG leaves and IG-equivalence). *The information–geometric comparison of realizations is organized by an equivalence relation $\cong_{\text{IG}}$ on the realization space $\mathcal{R}(\Gamma_Y)$, whose equivalence classes (“IG leaves”) partition $\mathcal{R}(\Gamma_Y)$. We give the formal definitions later in §3.3 (Definitions 22 and 23). For the boundary-first development here, it suffices to remember that IG leaves encode “mutually intelligible” interior interpretations that share the same boundary record at the chosen resolution.*

Remark 4 (Fibre intersects many leaves). *The fibre $\text{Fib}(\Gamma_Y)$ is a boundary constraint (boundary-probability level), while IG leaves are internal geometric identifications (information-geometric level). In general,*

$$\text{Fib}(\Gamma_Y) \text{ intersects many IG leaves.}$$

Our selection problem is therefore: find (or characterize) the IG leaves whose representatives lie in (or near) $\text{Fib}(\Gamma_Y)$.

1.3 From an edge law to a step operator (boundary-probability level)

Given a derived edge record $\Gamma_Y^{(\Delta t)} \in \mathcal{P}(Y \times Y)$, if it admits a disintegration $\Gamma_Y(dy_0, dy_1) = \mu(dy_0) K(y_0, dy_1)$, then K defines a one-step Markov operator $Tf(y) := \int f(y') K(y, dy')$ on a chosen function space $\mathcal{X}$. We treat this as *derived structure*: disintegration is a property of $\Gamma_Y^{(\Delta t)}$ relative to a reference marginal μ (and depends on the chosen lag Δt).

1.4 Semigroup consistency and generator limits (boundary-probability level)

Definition 6 (boundary-probability level: empirical semigroup consistency). *Let $\{T_{\Delta t}\}_{\Delta t \in (0, \Delta t_0]}$ be operators on a Banach space $\mathcal{X}$ extracted from boundary records at different resolutions (e.g. from $\Gamma_Y^{(\Delta t)}$). We say the family is empirically semigroup-consistent if:*

- (MC1) **Approximate semigroup:** $\|T_{t+s} - T_t T_s\|_{\mathcal{B}(\mathcal{X})} \rightarrow 0$ as $\max\{t, s\} \rightarrow 0$ for rational t, s with $t + s \leq \Delta t_0$;
- (MC2) **Continuity at 0:** $T_0 = I$ and $\|T_{\Delta t} - I\|_{\mathcal{B}(\mathcal{X})} \rightarrow 0$ as $\Delta t \rightarrow 0$;
- (MC3) **Generator tightness:** the difference quotients $L_{\Delta t} := (T_{\Delta t} - I)/\Delta t$ are dissipative and satisfy the standard Hille–Yosida type bounds needed for a contraction semigroup limit (when it exists).

Proposition 2 (boundary-probability level: existence of a continuous-time Markov completion (closure form)). *Assume Definition 6 and suppose $L_{\Delta t} \rightarrow L$ in strong resolvent sense to a closed, densely defined operator L on $\mathcal{X}$. Then there exists a strongly continuous contraction semigroup $(P_t)_{t \geq 0}$ on $\mathcal{X}$ with generator L such that*

$$P_{\Delta t} f = \lim_{n \rightarrow \infty} (T_{\Delta t/n})^n f, \quad f \in \mathcal{X}.$$

In particular, a continuous-time Markov completion is obtained as a closure of the boundary-derived skeleton, not as a primitive assumption.

Remark 5 (Design choice: completion target class). *The “biconditional” language in the prose should be read as an existence/fixed-point statement relative to a chosen target class: given a boundary record and a class of admissible semigroups (e.g. Feller, diffusion, jump processes), completion means showing that the boundary-derived operators admit a closure within that class. Different target classes lead to different notions of completion.*

2 Neutral operator framework (technical backbone)

Roadmap. This section defines the neutral operator as a boundary-factorized functional and isolates the design choice of a completion target class. The key message is that any Markov semigroup structure enters only through stated approximation/limit hypotheses (semigroup consistency, generator convergence), not by stipulation. This note collects the core definitions, theorems, and relations used in Part I, without narrative. The standing objects are:

- a measurable observation space Y ;
- an observed bi-infinite time series $(Y_t)_{t \in \mathbb{Z}}$ taking values in Y ;
- an empirical one-step marginal $\Gamma_Y \in \mathcal{P}(Y \times Y)$, estimated from data.

In this section we shift from boundary-first compatibility (path-level horizons) to the information–geometric comparison of interior realizations. Since the IG structure is carried by a *finite-dimensional* boundary statistic, we implicitly fix a lag/resolution Δt^* and work with the derived edge law $\Gamma_Y := \Gamma_Y^{(\Delta t^*)}$. When needed (e.g. for completion), we consider a family of lags H and the corresponding collection $\{\Gamma_Y^{(\Delta t)}\}_{\Delta t \in H}$ as derived shadows of the same boundary horizon.

1. Empirical, representation, and latent operators

Definition 7 (Boundary-readable realization). *Fix a boundary horizon $\mathbb{P}_Y \in \mathcal{P}(Y^{\mathbb{Z}})$ and (when needed) a lag Δt^* with derived edge record $\Gamma_Y := \Gamma_Y^{(\Delta t^*)} \in \mathcal{P}(Y \times Y)$. A boundary-readable realization of Γ_Y is a triple (S, Γ_S, q) where S is a measurable space, $q : S \rightarrow Y$ is a measurable readout, and $\Gamma_S \in \mathcal{P}(S \times S)$ is a stepwise law such that*

$$(q \otimes q)_\# \Gamma_S = \Gamma_Y.$$

We write $\partial(S, \Gamma_S, q) := (q \otimes q)_\# \Gamma_S$ for the induced boundary map.

Lemma 1 (Pushforward-only comparison on a representation boundary). *Let $\Gamma_Y \in \mathcal{P}(Y \times Y)$ be a boundary record and let $\rho : Y \rightarrow Z$ be any measurable representation statistic. For any candidate boundary law $\Gamma'_Y \in \mathcal{P}(Y \times Y)$ (for example, $\Gamma'_Y = (h \times h)_\# \Gamma_X$ coming from a latent realization (X, Γ_X, h)), define the induced representation-boundary records*

$$\Gamma_{Y,\rho} := (\rho \times \rho)_\# \Gamma_Y \in \mathcal{P}(Z \times Z), \quad \Gamma_{Y',\rho} := (\rho \times \rho)_\# \Gamma'_Y \in \mathcal{P}(Z \times Z).$$

Then any divergence D_f on probability laws may be evaluated entirely at the representation boundary via

$$D_f(\Gamma_{Y,\rho} \parallel \Gamma_{Y',\rho}),$$

without requiring an inverse map $(\rho \times \rho)^{-1}$ or a decoder $Z \rightarrow Y$.

Lemma 2 (Invariance under measurable isomorphisms). *Let (S, Γ_S, q) be a realization in $\mathcal{R}(\Gamma_Y)$ and let $\psi : S \rightarrow \tilde{S}$ be a bimeasurable bijection with measurable inverse. Define*

$$\tilde{\Gamma} := (\psi \times \psi)_\# \Gamma_S, \quad \tilde{q} := q \circ \psi^{-1}.$$

Then $(\tilde{S}, \tilde{\Gamma}, \tilde{q})$ is again a realization in $\mathcal{R}(\Gamma_Y)$, and $(S, \Gamma_S, q) \cong_{\text{IG}} (\tilde{S}, \tilde{\Gamma}, \tilde{q})$. In particular, IG leaves and the neutral operator are invariant under measurable relabellings of state spaces.

Proof. First, $(\tilde{S}, \tilde{\Gamma}, \tilde{q}) \in \mathcal{R}(\Gamma_Y)$. Since (S, Γ_S, q) is a realization, we have

$$(q \times q)_\# \Gamma_S = \Gamma_Y.$$

Using functoriality of pushforwards and $\tilde{q} \circ \psi = q$, we compute

$$(\tilde{q} \times \tilde{q})_\# \tilde{\Gamma} = ((q \circ \psi^{-1}) \times (q \circ \psi^{-1}))_\# (\psi \times \psi)_\# \Gamma_S = (q \times q)_\# \Gamma_S = \Gamma_Y,$$

so $(\tilde{S}, \tilde{\Gamma}, \tilde{q})$ is a realization of Γ_Y .

For IG-equivalence, choose a disintegration $\Gamma_S(ds_0, ds_1) = \nu_S(ds_0) K_S(s_0, ds_1)$ and define the corresponding disintegration of $\tilde{\Gamma}$ by transport along ψ :

$$\tilde{\nu} := \psi_\# \nu_S, \quad \tilde{K}(\tilde{s}_0, \cdot) := (\psi)_\# (K_S(\psi^{-1}(\tilde{s}_0), \cdot)).$$

Then the conditional-law maps into $\mathcal{P}(Y)$ satisfy

$$\Phi_{\tilde{S}}(\tilde{s}) := \mathbb{P}(Y_{t+1} \in \cdot \mid \tilde{S}_t = \tilde{s}) = \mathbb{P}(Y_{t+1} \in \cdot \mid S_t = \psi^{-1}(\tilde{s})) = \Phi_S(\psi^{-1}(\tilde{s})) = \Phi_S \circ \psi^{-1}(\tilde{s}).$$

In particular, $\Phi_{\tilde{S}}(\tilde{S}) = \Phi_S(S)$, so the image statistical manifold $\mathcal{M}_0 \subset \mathcal{P}(Y)$ is the same for both realizations.

Because ψ is a measurable bijection with measurable inverse, the pullback of the Fisher metric and dual connections along ψ is an isomorphism of dualistic manifolds:

$$g_{\tilde{S}} = \psi_\# g_S, \quad \nabla^{\tilde{S}} = \psi_\# \nabla^S, \quad (\nabla^{\tilde{S}})^* = \psi_\# (\nabla^S)^*.$$

Equivalently, taking $S_0 = S$ and $\tilde{S}_0 = \tilde{S}$, the map $\varphi := \psi^{-1} : \tilde{S}_0 \rightarrow S_0$ satisfies the hypotheses of the IG-embedding theorem, and shows that $(S, \Gamma_S, q) \cong_{\text{IG}} (\tilde{S}, \tilde{\Gamma}, \tilde{q})$. $\square$

2. IG structure on $\mathcal{P}(Y \times Y)$

On any smooth statistical model $\theta \mapsto \Gamma_\theta \in \mathcal{P}(Y \times Y)$, the second variation of D_f at θ yields:

Definition 8 (Fisher–Rao metric and dual connections). *Let $\{\Gamma_\theta : \theta \in \Theta\}$ be a smooth statistical model. The Riemannian metric g on Θ with local components*

$$g_{ij}(\theta) := \left. \frac{\partial^2}{\partial \epsilon_i \partial \epsilon_j} D_f(\Gamma_\theta \| \Gamma_{\theta+\epsilon}) \right|_{\epsilon=0}$$

is the Fisher–Rao metric. There exist two affine connections ∇ and ∇^ , dual with respect to g , such that for all vector fields X, Y, Z on Θ ,*

$$Y \langle X, Z \rangle = \langle \nabla_Y X, Z \rangle + \langle X, \nabla_Y^* Z \rangle.$$

The triple (g, ∇, ∇^) is the dualistic structure (independent of f up to scale, for regular Csiszár–Amari α -divergences).*

3. Conditional-law maps and Fisher immersions

Given $\Gamma_Y \in \mathcal{P}(Y \times Y)$, choose a disintegration

$$\Gamma_Y(dy_0, dy_1) = \nu_Y(dy_0) K_Y(y_0, dy_1),$$

with source law ν_Y and Markov kernel K_Y .

Definition 9 (Conditional-law map). For a realization $(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$, with disintegration

$$\Gamma_S(ds_0, ds_1) = \nu_S(ds_0) K_S(s_0, ds_1),$$

the conditional-law map with respect to Y is

$$\Phi_S : S \rightarrow \mathcal{P}(Y), \quad \Phi_S(s) := \mathbb{P}(Y_{t+1} \in \cdot \mid S_t = s).$$

When $S = Y$ and $q = \text{id}_Y$, this reduces to $y \mapsto K_Y(y, \cdot)$.

Pulling back the dualistic structure along Φ_S gives:

Definition 10 (Induced IG structure on a realization). Let (g, ∇, ∇^*) be the dualistic structure on a smooth submanifold $\mathcal{M} \subset \mathcal{P}(Y)$ containing the image of Φ_S . The induced IG structure on S is

$$g_S := \Phi_S^* g, \quad \nabla^S := \Phi_S^* \nabla, \quad (\nabla^S)^* := \Phi_S^* \nabla^*.$$

Definition 11 (Fisher immersion). Let S be a smooth manifold and $\Phi_S : S \rightarrow \mathcal{M} \subset \mathcal{P}(Y)$ smooth. We say Φ_S is a Fisher immersion of rank d on a subset $S_0 \subseteq S$ if the pullback metric g_S has constant rank d on S_0 and the induced map $\Phi_S|_{S_0}$ is an immersion of S_0 into $\mathcal{M}$.

4. IG embedding and IG-equivalence

Theorem 1 (IG-embedding theorem). Let (X, Γ_X, h) be a latent hypothesis and (Z, Γ_Z, r) a representation, both in $\mathcal{R}(\Gamma_Y)$, such that:

1. **Rose condition:** the pushforward of Γ_X satisfies

$$(h \times h)_\# \Gamma_X = \Gamma_Y.$$

2. **Fisher immersions:** there exist subsets $X_0 \subseteq X$, $Z_0 \subseteq Z$ and a common statistical manifold $\mathcal{M}_0 \subset \mathcal{P}(Y)$ such that

$$\Phi_X(X_0) = \Phi_Z(Z_0) = \mathcal{M}_0,$$

and $\Phi_X|_{X_0}, \Phi_Z|_{Z_0}$ are Fisher immersions of the same rank d .

Then:

- there is a canonical smooth coordinate changes

$$\varphi := (\Phi_X|_{X_0})^{-1} \circ (\Phi_Z|_{Z_0}) : Z_0 \rightarrow X_0;$$

- φ is an isomorphism of dualistic manifolds:

$$g_Z = \varphi^* g_X, \quad \nabla^Z = \varphi^* \nabla^X, \quad (\nabla^Z)^* = \varphi^* (\nabla^X)^*;$$

- φ intertwines the stepwise marginals on the supports:

$$\Gamma_Z|_{Z_0 \times Z_0} = (\varphi^{-1} \times \varphi^{-1})_\# (\Gamma_X|_{X_0 \times X_0}).$$

Definition 12 (IG-equivalence). Two realizations (S, Γ_S, q) and $(S', \Gamma_{S'}, q')$ in $\mathcal{R}(\Gamma_Y)$ are information-geometrically equivalent (IG-equivalent), written

$$(S, \Gamma_S, q) \cong_{\text{IG}} (S', \Gamma_{S'}, q'),$$

if there exist subsets $S_0 \subseteq S$, $S'_0 \subseteq S'$ and a common statistical manifold $\mathcal{M}_0 \subset \mathcal{P}(Y)$ such that:

- $\Phi_S(S_0) = \Phi_{S'}(S'_0) = \mathcal{M}_0$;
- $\Phi_S|_{S_0}$ and $\Phi_{S'}|_{S'_0}$ are Fisher immersions of the same rank;
- the induced map $\varphi := (\Phi_S|_{S_0})^{-1} \circ (\Phi_{S'}|_{S'_0})$ is a dualistic isomorphism and intertwines the restricted stepwise marginals as above.

Definition 13 (IG leaf and foliation). *For any realization $(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$, the information-geometric leaf (IG leaf) through (S, Γ_S, q) is the IG-equivalence class*

$$[S, \Gamma_S, q]_{\text{IG}} := \{(S', \Gamma_{S'}, q') \in \mathcal{R}(\Gamma_Y) \mid (S', \Gamma_{S'}, q') \cong_{\text{IG}} (S, \Gamma_S, q)\}.$$

Assuming $\mathcal{R}(\Gamma_Y)$ carries a smooth (or stratified) manifold structure and the constructions above depend smoothly on (S, Γ_S, q) , the family of leaves

$$\{[S, \Gamma_S, q]_{\text{IG}} : (S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)\}$$

partitions $\mathcal{R}(\Gamma_Y)$ into disjoint immersed submanifolds (leaves). When this partition satisfies the usual local product condition, it defines a foliation of $\mathcal{R}(\Gamma_Y)$ by IG leaves.

Definition 14 (Neutral operator). *Fix an empirical stepwise marginal $\Gamma_Y \in \mathcal{P}(Y \times Y)$ (e.g. $\Gamma_Y = \Gamma_Y^{(\Delta t^*)}$ derived from a boundary horizon as in Definition 1) and a regular Csiszár–Amari f -divergence D_f on $\mathcal{P}(Y \times Y)$. The associated neutral operator is the triple*

$$\mathcal{N}(\Gamma_Y, D_f) := (\Gamma_Y, \mathcal{R}(\Gamma_Y), \cong_{\text{IG}}),$$

where $\mathcal{R}(\Gamma_Y)$ is the realization space of Γ_Y and $\cong_{\text{IG}}$ is IG-equivalence on $\mathcal{R}(\Gamma_Y)$ induced by D_f . The IG-equivalence classes $\{[S, \Gamma_S, q]_{\text{IG}}\}$ are called the information-geometric leaves (IG leaves) of the neutral operator.

2.1 Markov completion of a stepwise marginal

We now make explicit the sense in which a stepwise marginal $\Gamma_Y \in \mathcal{P}(Y \times Y)$ canonically determines a discrete-time Markov representation, while keeping this representation conceptually separate from the neutral operator $\mathcal{N}(\Gamma_Y, D_f)$.

By disintegration there exist a *source law* $\nu_Y \in \mathcal{P}(Y)$ and a Markov kernel $K_Y : Y \times \mathcal{B}(Y) \rightarrow [0, 1]$ such that

$$\Gamma_Y(dy_0, dy_1) = \nu_Y(dy_0) K_Y(y_0, dy_1). \quad (1)$$

The pair (ν_Y, K_Y) is unique up to ν_Y -null sets on Y .

Definition 15 (Discrete-time Markov completion of Γ_Y). *A discrete-time Markov completion of the stepwise marginal Γ_Y is the stationary Markov chain $(Y_t)_{t \in \mathbb{Z}}$ on Y with initial law $Y_0 \sim \nu_Y$ and transition kernel K_Y as in (1), i.e.*

$$\mathbb{P}(Y_{t+1} \in A \mid Y_t = y) = K_Y(y, A), \quad \mathbb{P}(Y_t \in A) = \nu_Y(A)$$

for all $A \in \mathcal{B}(Y)$ and $t \in \mathbb{Z}$. Equivalently, the lag-1 joint law of (Y_t) is precisely Γ_Y :

$$\mathbb{P}(Y_t \in A, Y_{t+1} \in B) = \Gamma_Y(A \times B).$$

Proposition 3 (Existence and uniqueness up to versions). *For every $\Gamma_Y \in \mathcal{P}(Y \times Y)$ there exists a discrete-time Markov completion in the sense of Definition 15, and it is unique up to versions of the disintegration (1).*

Proof. Existence and uniqueness of the disintegration $\Gamma_Y(dy_0, dy_1) = \nu_Y(dy_0)K_Y(y_0, dy_1)$ are standard under mild regularity hypotheses on $(Y, \mathcal{B}(Y))$ (e.g. standard Borel space). The Kolmogorov extension theorem then yields a unique (in law) stationary Markov chain with one-step kernel K_Y and invariant law ν_Y . Any other stationary Markov chain whose lag-1 joint law is Γ_Y must admit a disintegration of the same form, hence coincides with this completion up to versions of (ν_Y, K_Y) on ν_Y -null sets. $\square$

Remark 6 (Static nature of the neutral operator). *The neutral operator $\mathcal{N}(\Gamma_Y, D_f)$ from Definition 14 is a static object: it organizes realizations of the edge marginal Γ_Y and their information-geometric relations, independently of any particular choice of Markov chain or path-space dynamics. The discrete-time Markov completion (Y_t) is one convenient representation of Γ_Y , but it is not singled out ontologically by the neutral operator; it is merely a canonical way to extend the edge marginal to a path measure.*

2.2 Continuous-time Markov completions from empirical limits

We now describe how a continuous-time Markov semigroup can arise as a *completion* of an empirically observed family of stepwise marginals at different time scales, rather than being postulated a priori.

Definition 16 (Empirical edge-marginal family). *An empirical edge-marginal family on Y is a collection*

$$\{\Gamma_Y^{(\Delta t)}\}_{\Delta t \in (0, \Delta t_0]} \subset \mathcal{P}(Y \times Y)$$

such that for each Δt ,

$$\Gamma_Y^{(\Delta t)}(dy_0, dy_1) = \nu_Y(dy_0) K_Y^{(\Delta t)}(y_0, dy_1)$$

for a common source law ν_Y and some Markov kernel $K_Y^{(\Delta t)}$ on Y .

Let $\mathcal{X}$ denote a Banach space of observables on Y (e.g. bounded measurable functions, or a suitable L^p space with respect to ν_Y). Each kernel $K_Y^{(\Delta t)}$ induces a bounded linear operator $T_{\Delta t} : \mathcal{X} \rightarrow \mathcal{X}$ by

$$T_{\Delta t}f(y) := \int f(y') K_Y^{(\Delta t)}(y, dy').$$

Definition 17 (Empirical semigroup consistency). *We say that an empirical edge-marginal family $\{\Gamma_Y^{(\Delta t)}\}$ is empirically semigroup-consistent on $\mathcal{X}$ if the associated operators $\{T_{\Delta t}\}$ satisfy:*

(MC1) **Approximate semigroup property:** *For rational times $t, s > 0$ with $t, s, t + s \leq \Delta t_0$, we have*

$$\|T_{t+s} - T_t T_s\|_{\mathcal{B}(\mathcal{X})} \xrightarrow{\max\{t, s\} \rightarrow 0} 0.$$

(MC2) **Continuity at 0:** $T_0 := I$ on $\mathcal{X}$, and

$$\|T_{\Delta t} - I\|_{\mathcal{B}(\mathcal{X})} \xrightarrow{\Delta t \rightarrow 0} 0.$$

(MC3) **Uniform boundedness and dissipativity:** *The scaled difference operators*

$$L_{\Delta t} := \frac{T_{\Delta t} - I}{\Delta t}$$

are uniformly bounded and satisfy the usual positivity / contraction properties that guarantee, via Hille–Yosida, that their limit (if it exists) is the generator of a Markov semigroup on $\mathcal{X}$.

Proposition 4 (Existence of a continuous-time Markov completion). *Let $\{\Gamma_Y^{(\Delta t)}\}$ be an empirical edge-marginal family that is empirically semigroup-consistent on $\mathcal{X}$ in the sense of Definition 17. Suppose that $L_{\Delta t}$ converges in the strong operator topology on $\mathcal{X}$ to a closed, densely defined operator L :*

$$L_{\Delta t}f \xrightarrow{\Delta t \rightarrow 0} Lf, \quad f \in \mathcal{D}(L).$$

Then there exists a strongly continuous Markov semigroup $(P_t)_{t \geq 0}$ on $\mathcal{X}$ with generator L , such that

$$P_{\Delta t}f = \lim_{n \rightarrow \infty} (T_{\Delta t/n})^n f$$

for all $f \in \mathcal{X}$ and $\Delta t \in (0, \Delta t_0]$. In particular, the operators $T_{\Delta t}$ are approximations of the skeleton $\{P_{\Delta t}\}$ of the continuous-time semigroup.

Sketch of proof. Under (MC1)–(MC3), the family $\{T_{\Delta t}\}$ defines an approximate semigroup of contractions on $\mathcal{X}$ which is uniformly bounded and strongly continuous at 0. The convergence of $L_{\Delta t}$ to a closed, densely defined operator L satisfying the Hille–Yosida conditions implies that L is the generator of a unique strongly continuous contraction semigroup $(P_t)_{t \geq 0}$ on $\mathcal{X}$. Standard Trotter–Kato approximation theorems then yield the convergence

$$(T_{\Delta t/n})^n f \rightarrow P_{\Delta t}f$$

for each fixed Δt and $f \in \mathcal{X}$. □

Definition 18 (Continuous-time Markov completion). *Any strongly continuous Markov semigroup $(P_t)_{t \geq 0}$ on $\mathcal{X}$ obtained as in Proposition 4 is called a continuous-time Markov completion of the empirical edge-marginal family $\{\Gamma_Y^{(\Delta t)}\}$. When, in addition, the conditional-law manifold $\mathcal{M}_0 \subset \mathcal{P}(Y)$ is equipped with the canonical dualistic structure (g, ∇, ∇^*) from the canonical Fisher structure on $\mathcal{M}_0$, the semigroup (P_t) induces a Markov semigroup (P_t^*) on $\mathcal{M}_0$ compatible with this information-geometric structure.*

Remark 7 (From neutral operator to dynamical neutral operator). *In what follows, a dynamical neutral operator will always be understood as a neutral operator $\mathcal{N}(\Gamma_Y, D_f)$ together with a choice of continuous-time Markov completion $(P_t^*)_{t \geq 0}$ on the canonical conditional-law manifold $\mathcal{M}_0$ associated to Γ_Y . Different Markov completions that agree on the induced semigroup on $\mathcal{M}_0$ (up to IG-equivalence) live in the same dynamical neutral operator in the sense of the next subsection.*

5. Fidelity notions (Rose–Fisher ladder)

Measure fidelity (Rose condition).

Definition 19 (Edge divergence and Rose condition). *Let D_f be a smooth f -divergence on $\mathcal{P}(Y \times Y)$. For a candidate factorization $\Gamma_Z \in \mathcal{P}(Y \times Y)$, define the edge divergence*

$$\mathcal{A}_f^Y(\Gamma_Z) := D_f(\Gamma_Y \parallel \Gamma_Z).$$

We say that Γ_Z satisfies the Rose condition if

$$\mathcal{A}_f^Y(\Gamma_Z) = 0.$$

Under the usual domination assumptions this is equivalent to $\Gamma_Z = \Gamma_Y$ almost everywhere, i.e. exact one-step measure conjugacy.

Geometric fidelity (separability vs intrinsic coupling). Let $(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$ with induced IG structure $(g_S, \nabla^S, (\nabla^S)^*)$, and let $S \cong S^A \times S^B$ be a product decomposition of coordinates (e.g. two sensor groups, delay blocks, or latent/observed splits).

Definition 20 (Geometric fidelity across a cut). *We say that the cut (S^A, S^B) has geometric fidelity if there exist local coordinates (θ^A, θ^B) on S such that:*

(G1) (metric separability) g_S is block-diagonal:

$$g_S = g^A \oplus g^B;$$

(G2) (dual-connection separability) the Christoffel symbols of ∇^S and $(\nabla^S)^*$ have no mixed components in these coordinates, i.e. they split as products of connections on S^A and S^B .

In this case the dualistic structure factors as a product of two Fisher subsystems on S^A and S^B .

Definition 21 (Intrinsic coupling). *If no local reparameterization of S can be found that simultaneously satisfies (G1) and (G2) for the cut (S^A, S^B) , we say that the cut exhibits intrinsic coupling. In this case the joint information geometry contains irreducible cross-terms that cannot be removed within the IG leaf.*

Remark 8 (Rose–Fisher ladder). *Topological fidelity (via Fisher immersions), measure fidelity (Rose condition), and geometric fidelity (separability vs intrinsic coupling) form a three-level hierarchy:*

$$\text{State-space (optional)} \Rightarrow \text{Measure} \Rightarrow \text{Geometric},$$

in the sense that each level adds additional structure on top of Γ_Y while remaining neutral with respect to the choice of Csiszár–Amari α -divergence.

3 Core results of the neutral operator framework

Roadmap. Here we collect the main structural statements used later: how neutrality forces dependence only on the boundary record, how fibres organise admissible interiors, and how completion classes behave under the boundary map. These results are the formal bridge between the boundary-first definitions and the empirical/algorithmic constructions. In this section we collect the main structural results used throughout. We assume the background from Section B: stepwise marginals $\Gamma_S \in \mathcal{P}(S \times S)$, Csiszár–Morimoto f -divergences, Fisher–Rao metrics and dual connections, and the realization space $\mathcal{R}(\Gamma_Y)$ and IG-equivalence defined elsewhere.

3.1 Rose condition and transport equivalence

We start by making precise the sense in which the Rose condition captures equality of one-step transport.

Theorem 2 (Rose condition and equality of stepwise marginals). *Let D_f be a Csiszár–Morimoto f -divergence on $\mathcal{P}(Y \times Y)$ induced by a strictly convex function $f : (0, \infty) \rightarrow \mathbb{R}$ with $f(1) = 0$. Let $\Gamma_Y, \Gamma_Z \in \mathcal{P}(Y \times Y)$ be two stepwise marginals such that $\Gamma_Y \ll \lambda$ and $\Gamma_Z \ll \lambda$ for some common reference measure λ on $Y \times Y$, and write $\rho := \frac{d\Gamma_Y}{d\Gamma_Z}$ whenever $\Gamma_Y \ll \Gamma_Z$.*

Then:

(a) *If $D_f(\Gamma_Y \| \Gamma_Z) = 0$, then $\Gamma_Y = \Gamma_Z$ λ -almost everywhere.*

(b) Conversely, if $\Gamma_Y = \Gamma_Z$ λ -almost everywhere, then $D_f(\Gamma_Y \parallel \Gamma_Z) = 0$.

In particular, for any such D_f the Rose condition

$$D_f(\Gamma_Y \parallel \Gamma_Z) = 0$$

is equivalent to equality of stepwise marginals, and hence to equality of the associated one-step transport (up to null sets).

Proof. If $\Gamma_Y \not\ll \Gamma_Z$ then $D_f(\Gamma_Y \parallel \Gamma_Z) = +\infty$ by definition, so $D_f(\Gamma_Y \parallel \Gamma_Z) = 0$ implies $\Gamma_Y \ll \Gamma_Z$. In this case,

$$D_f(\Gamma_Y \parallel \Gamma_Z) = \int f(\rho) \Gamma_Z(dy_0, dy_1), \quad \rho := \frac{d\Gamma_Y}{d\Gamma_Z}.$$

Strict convexity of f with $f(1) = 0$ implies $f(x) \geq 0$ with equality if and only if $x = 1$. Thus $D_f(\Gamma_Y \parallel \Gamma_Z) = 0$ forces $f(\rho) = 0$ Γ_Z -a.e., hence $\rho = 1$ Γ_Z -a.e., i.e. $\Gamma_Y = \Gamma_Z$ λ -a.e. The converse is immediate from the definition. $\square$

Throughout we use the shorthand *Rose condition* for the equality $D_f(\Gamma_Y \parallel \Gamma_Z) = 0$ for a chosen regular f -divergence.

Corollary 1 (Conditional-law equivalence for realizations). *Let (S, Γ_S, q) and $(S', \Gamma_{S'}, q')$ be realizations in $\mathcal{R}(\Gamma_Y)$ with disintegrations*

$$\Gamma_S(ds_0, ds_1) = \nu_S(ds_0) K_S(s_0, ds_1), \quad \Gamma_{S'}(ds'_0, ds'_1) = \nu_{S'}(ds'_0) K_{S'}(s'_0, ds'_1).$$

Assume each realization pushes forward to the same empirical operator:

$$(q \times q)_\# \Gamma_S = (q' \times q')_\# \Gamma_{S'} = \Gamma_Y.$$

Then the following are equivalent:

- (i) $\Gamma_S = \Gamma_{S'}$ up to null sets on $S \times S$;
- (ii) $(\nu_S, K_S) = (\nu_{S'}, K_{S'})$ up to null sets on S and S' ;
- (iii) the conditional-law maps Φ_S and $\Phi_{S'}$ into $\mathcal{P}(Y)$ coincide almost everywhere with respect to the corresponding source laws.

3.2 Neutrality of α -divergences for IG structure

We next record the sense in which Csiszár–Amari α -divergences are *neutral* with respect to the induced information geometry.

Theorem 3 (α -neutrality of IG structure). *Let $\{\Gamma_\theta : \theta \in \Theta\}$ be a smooth statistical model of stepwise marginals on $Y \times Y$, with corresponding parameter manifold Θ . For each α in the regular range, let D_α be the Csiszár–Amari α -divergence on $\mathcal{P}(Y \times Y)$, and let $(g^{(\alpha)}, \nabla^{(\alpha)}, (\nabla^{(\alpha)})^*)$ denote the metric and dual affine connections induced on Θ by D_α .*

Then:

- (a) For any α, β in the regular range and any $\Gamma_Y, \Gamma_Z \in \mathcal{P}(Y \times Y)$,

$$D_\alpha(\Gamma_Y \parallel \Gamma_Z) = 0 \iff D_\beta(\Gamma_Y \parallel \Gamma_Z) = 0,$$

and both are equivalent to $\Gamma_Y = \Gamma_Z$ almost everywhere.

- (b) The metrics $g^{(\alpha)}$ and $g^{(\beta)}$ are proportional: there exists a constant $c_{\alpha,\beta} > 0$ such that $g^{(\alpha)} = c_{\alpha,\beta} g^{(\beta)}$.
- (c) The dualistic structures $(g^{(\alpha)}, \nabla^{(\alpha)}, (\nabla^{(\alpha)})^*)$ and $(g^{(\beta)}, \nabla^{(\beta)}, (\nabla^{(\beta)})^*)$ are equivalent up to this overall scaling and smooth reparametrization of Θ .

In particular, IG-equivalence classes and IG leaves in the realization space $\mathcal{R}(\Gamma_Y)$ do not depend on the choice of α .

Proof sketch. Part (a) follows from Theorem 2 applied to each D_α separately, since each D_α is itself an f -divergence generated by a strictly convex f_α with $f_\alpha(1) = 0$.

For (b) and (c), it is a standard result of information geometry that the Fisher–Rao metric is essentially the unique monotone Riemannian metric on a statistical manifold (Čencov’s theorem), and that the family of α -divergences all induce, up to a constant factor, the same Fisher metric and associated dualistic structure on Θ ; see, e.g., Amari [?] for a detailed treatment. We simply record this neutrality in the present notation. $\square$

3.3 IG-equivalence, IG leaves, and the neutral operator

We now formalize the information–geometric equivalence between realizations and package it into the neutral operator. Throughout we fix a regular Csiszár–Morimoto f -divergence D_f on $\mathcal{P}(Y \times Y)$, and let (g, ∇, ∇^*) denote the dualistic structure induced on smooth statistical models by D_f as in Definition 44.

By Čencov’s characterization of monotone metrics (together with the α -neutrality of Theorem 3), the Fisher metric and associated dual affine connections are *canonical* up to an overall constant factor among Riemannian metrics that are monotone under Markov morphisms. In particular, once D_f is fixed, the induced IG structure on any conditional-law manifold is unique up to scale.

Theorem 4 (Information–geometric embedding). *Let (X, Γ_X, h) and (Z, Γ_Z, r) be realizations in $\mathcal{R}(\Gamma_Y)$, with*

$$\Gamma_Y = (h \times h)_\# \Gamma_X, \quad \Gamma_Z = (r \times r)_\# \Gamma_Y.$$

Let $\Phi_X : X \rightarrow \mathcal{P}(Y)$ and $\Phi_Z : Z \rightarrow \mathcal{P}(Y)$ be the associated conditional-law maps from Definition 9, and let D_f induce a Fisher metric g on their common image.

Assume there exist subsets $X_0 \subseteq X$ and $Z_0 \subseteq Z$ and a statistical manifold $\mathcal{M}_0 \subseteq \mathcal{P}(Y)$ such that:

- (E1) $\Phi_X(X_0) = \Phi_Z(Z_0) = \mathcal{M}_0$;
- (E2) *the restricted maps $\Phi_X|_{X_0} : X_0 \rightarrow \mathcal{M}_0$ and $\Phi_Z|_{Z_0} : Z_0 \rightarrow \mathcal{M}_0$ are Fisher immersions of the same rank d .*

Then:

- (a) *There is a canonical smooth map*

$$\varphi := (\Phi_X|_{X_0})^{-1} \circ (\Phi_Z|_{Z_0}) : Z_0 \rightarrow X_0$$

which is a smooth coordinate changes onto its image, with smooth inverse on each connected component supporting the dynamics.

(b) The map φ is a dualistic isomorphism: it is a Fisher isometry between (Z_0, g_{Z_0}) and (X_0, g_{X_0}) and intertwines the dual connections induced by D_f :

$$g_{Z_0} = \varphi^* g_{X_0}, \quad \nabla^{Z_0} = \varphi^* \nabla^{X_0}, \quad (\nabla^{Z_0})^* = \varphi^* (\nabla^{X_0})^*.$$

(c) The restricted stepwise marginals are related by pushforward:

$$\Gamma_Z|_{Z_0 \times Z_0} = (\varphi^{-1} \times \varphi^{-1})_\# (\Gamma_X|_{X_0 \times X_0}).$$

In this situation we say that (Z, Γ_Z, r) is an information–geometric embedding of (X, Γ_X, h) on (Z_0, X_0) .

Definition 22 (IG-equivalence). Two realizations (S, Γ_S, q) and $(S', \Gamma_{S'}, q')$ in $\mathcal{R}(\Gamma_Y)$ are information–geometrically equivalent (IG-equivalent), written

$$(S, \Gamma_S, q) \cong_{\text{IG}} (S', \Gamma_{S'}, q'),$$

if there exist subsets $S_0 \subseteq S$, $S'_0 \subseteq S'$ and a common statistical manifold $\mathcal{M}_0 \subseteq \mathcal{P}(Y)$ such that:

- $\Phi_S(S_0) = \Phi_{S'}(S'_0) = \mathcal{M}_0$;
- the restrictions $\Phi_S|_{S_0}$ and $\Phi_{S'}|_{S'_0}$ are Fisher immersions of the same rank;
- the induced map

$$\varphi := (\Phi_S|_{S_0})^{-1} \circ (\Phi_{S'}|_{S'_0}) : S'_0 \rightarrow S_0$$

is a dualistic isomorphism and intertwines the restricted stepwise marginals as in Theorem 4.

Definition 23 (IG leaf). For any realization $(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$, the information–geometric leaf (IG leaf) through (S, Γ_S, q) is the IG-equivalence class

$$[S, \Gamma_S, q]_{\text{IG}} := \{(S', \Gamma_{S'}, q') \in \mathcal{R}(\Gamma_Y) \mid (S', \Gamma_{S'}, q') \cong_{\text{IG}} (S, \Gamma_S, q)\}.$$

Proposition 5 (IG-equivalence is an equivalence relation). The relation $\cong_{\text{IG}}$ on $\mathcal{R}(\Gamma_Y)$ is an equivalence relation: it is reflexive, symmetric, and transitive (up to null sets on the relevant supports). Consequently, the IG leaves $\{[S, \Gamma_S, q]_{\text{IG}}\}$ form a partition of $\mathcal{R}(\Gamma_Y)$ into disjoint classes.

3.4 Foliation by IG leaves

Under additional regularity assumptions we can regard these IG-equivalence classes as leaves of a foliation.

Theorem 5 (Foliation by IG leaves). Suppose the realization space $\mathcal{R}(\Gamma_Y)$ admits the structure of a smooth (or stratified) manifold such that:

(F1) for each realization (S, Γ_S, q) , the IG-equivalence class

$$[S, \Gamma_S, q]_{\text{IG}} := \{(S', \Gamma_{S'}, q') \in \mathcal{R}(\Gamma_Y) \mid (S', \Gamma_{S'}, q') \cong_{\text{IG}} (S, \Gamma_S, q)\}$$

is an immersed connected submanifold of $\mathcal{R}(\Gamma_Y)$;

(F2) the dependence of the induced dualistic structure on (S, Γ_S, q) is smooth and the IG-equivalence classes vary smoothly in local charts.

Then the collection of IG leaves

$$\{[S, \Gamma_S, q]_{\text{IG}} : (S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)\}$$

defines a foliation of $\mathcal{R}(\Gamma_Y)$ in the sense of Definition 52.

Proof sketch. By Proposition 5 the sets $[S, \Gamma_S, q]_{\text{IG}}$ form a partition of $\mathcal{R}(\Gamma_Y)$ into disjoint subsets. Assumption (F1) identifies each equivalence class with an immersed connected submanifold (the candidate leaf), and (F2) ensures that in local coordinates the classes intersect coordinate patches in slices that are diffeomorphic to $\mathbb{R}^p \times \{y_0\}$ for some p, q , yielding the local product structure required by the definition of a foliation. $\square$

3.5 Geometric fidelity, empirical separability, and product structure

We now make precise how geometric fidelity across a cut is reflected in the product structure of the induced dualistic geometry, and how this yields a notion of empirical separability.

Proposition 6 (Geometric fidelity and product dualistic structure). *Let $(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$ be a realization with induced dualistic structure $(g_S, \nabla^S, (\nabla^S)^*)$ on a parameter manifold Θ_S . Suppose Θ_S admits a decomposition $\Theta_S \cong \Theta^A \times \Theta^B$ corresponding to a cut of the representation into two blocks (A, B) .*

Then the following are equivalent:

- (G1) *The cut (A, B) has geometric fidelity in the sense of Definition 20: there exist local coordinates (θ^A, θ^B) such that the Fisher metric is block-diagonal with respect to the cut and there are no mixed Fisher components between A and B .*
- (G2) *Locally, $(\Theta_S, g_S, \nabla^S, (\nabla^S)^*)$ is isomorphic, as a dualistic manifold, to the product*

$$(\Theta^A, g^A, \nabla^A, (\nabla^A)^*) \times (\Theta^B, g^B, \nabla^B, (\nabla^B)^*),$$

i.e. there exist local coordinates (θ^A, θ^B) in which

$$g_S = g^A \oplus g^B,$$

and the Christoffel symbols of ∇^S and $(\nabla^S)^$ have no mixed components between θ^A and θ^B .*

When these conditions hold we say that the cut (A, B) is geometrically separable. Otherwise we speak of intrinsic coupling across the cut.

Remark 9 (Dualistic interpretation of empirical separability). *Let $M_0 \subset \mathcal{P}(Y)$ be a conditional-law manifold carrying a dualistic structure (g, ∇, ∇^*) induced by some regular f -divergence. Each realization (S, Γ_S, q) with conditional-law map $\Phi_S : S \rightarrow M_0$ defines a distinguished subbundle*

$$H_\mu^S := d\Phi_S(T_s S) \subset T_\mu M_0, \quad \mu = \Phi_S(s),$$

consisting of directions in M_0 reachable by perturbing S .

For instance, a latent hypothesis (X, Γ_X, h) , the empirical realization $(Y, \Gamma_Y, \text{id}_Y)$, and a representation (Z, Γ_Z, r) induce subbundles

$$H^X, H^Y, H^Z \subset TM_0.$$

If we choose smooth vector fields

$$X \in \Gamma(H^X), \quad Y \in \Gamma(H^Y), \quad Z \in \Gamma(H^Z),$$

the duality relation

$$Y\langle X, Z \rangle = \langle \nabla_Y X, Z \rangle + \langle X, \nabla_Y^* Z \rangle$$

admits the following empirical reading:

- $\langle X, Z \rangle$ is the Fisher inner product between a latent direction and a representation direction at a fixed conditional law;
- $Y\langle X, Z \rangle$ measures how this inner product changes when we move along an observed direction;
- the two terms on the right-hand side describe how X and Z are “bent” along Y under the dual connections ∇ and ∇^* .

If, after reparameterization within an IG leaf, one can arrange that

$$\langle X, Z \rangle = \langle \nabla_Y X, Z \rangle = \langle X, \nabla_Y^* Z \rangle = 0$$

for all choices of $X \in H^X$, $Z \in H^Z$, $Y \in H^Y$ in some region, then the latent and representation subbundles remain orthogonal and invariant along observed directions: latent and representation directions do not shear into each other as we move along Y . This is exactly the situation of geometric separability.

If no such reparameterization exists inside the IG leaf, the cross terms cannot be removed, and we say that the corresponding split exhibits intrinsic coupling: the one-step empirical transport encodes an irreducible interaction between latent and representation directions along the observed manifold.

Definition 24 (Empirical separability across an observed cut). *Let $\Gamma_Y \in \mathcal{P}(Y \times Y)$ be fixed, and let $\mathcal{R}(\Gamma_Y)$ be its realization space. Consider a decomposition of the observation space $Y \cong Y^A \times Y^B$ (for instance, two sensor groups or coarse/fine coordinates), and let (A, B) denote the corresponding cut.*

We say that the cut (A, B) is empirically separable for Γ_Y if there exists at least one realization $(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$ and a decomposition $S \cong S^A \times S^B$ such that:

- (E1) *the map $q : S \rightarrow Y$ respects the cut, in the sense that $q(s^A, s^B) = (q^A(s^A), q^B(s^B))$ for measurable q^A, q^B ;*
- (E2) *the cut (S^A, S^B) is geometrically separable in the sense of Proposition 6.*

If no such realization and decomposition exist inside $\mathcal{R}(\Gamma_Y)$, we say that the cut (A, B) is empirically intrinsically coupled for Γ_Y .

Definition 25 (Empirical separability for (X, Y, Z)). *Let $\Gamma_Y \in \mathcal{P}(Y \times Y)$ be fixed, and let*

$$(X, \Gamma_X, h), \quad (Y, \Gamma_Y, \text{id}_Y), \quad (Z, \Gamma_Z, r)$$

be realizations in $\mathcal{R}(\Gamma_Y)$, with $h : X \rightarrow Y$ a latent hypothesis and $r : Y \rightarrow Z$ a representation.

We say that the triple (X, Y, Z) is empirically separable along the latent/observed/representation split if there exists a realization $(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$ and a product decomposition

$$S \cong S^X \times S^Y \times S^Z$$

such that:

- (ES1) the map $q : S \rightarrow Y$ factors coordinatewise, $q(s^X, s^Y, s^Z) = q^Y(s^Y)$, and h and r are obtained from S^X and S^Z via suitable measurable maps composed with q (as in the neutral diagram);
- (ES2) the induced dualistic structure on the parameter manifold for (S, Γ_S) is locally a product

$$(g_S, \nabla^S, (\nabla^S)^*) \cong (g^X, \nabla^X, (\nabla^X)^*) \times (g^Y, \nabla^Y, (\nabla^Y)^*) \times (g^Z, \nabla^Z, (\nabla^Z)^*),$$

i.e. there are local coordinates in which the Fisher metric and both dual connections have no mixed components across the three blocks.

If no such (S, Γ_S, q) and decomposition exist within $\mathcal{R}(\Gamma_Y)$, we say that the triple (X, Y, Z) is empirically intrinsically coupled along the latent/observed/representation split.

4 Information–geometric dynamical systems

Roadmap. We introduce the information-geometric structure needed to compare realizations beyond measure-level fidelity: statistical families over boundary records, the Fisher metric, dual affine connections, and projection geometry. This is where *IG leaves* become a concrete equivalence notion and where Pythagorean identities can be used as optional decompositions of boundary discrepancies. In this section we make explicit the level at which dynamics, information, and geometry meet. The basic objects are:

- a statistical manifold $\mathcal{M}_0 \subset \mathcal{P}(Y)$ of conditional laws;
- a dualistic structure (g, ∇, ∇^*) on $\mathcal{M}_0$ induced by a regular Csiszár–Amari f -divergence D_f ;
- a Markov semigroup $(P_t^*)_{t \geq 0}$ on $\mathcal{M}_0$ describing the time evolution of laws.

We first give an abstract definition, then specialize to the empirical setting associated with a fixed stepwise marginal Γ_Y and the neutral operator $\mathcal{N}(\Gamma_Y, D_f)$.

4.1 Dynamical neutral operator and IG–spectral embeddings

We now add dynamics to the neutral operator by choosing a continuous-time Markov completion on the canonical conditional-law manifold $\mathcal{M}_0$. Let $(P_t^*)_{t \geq 0}$ be a strongly continuous Markov semigroup acting on $\mathcal{M}_0 \subset \mathcal{P}(Y)$, with generator L and resolvent $R(\lambda)$ on a suitable Banach space of observables. For each realization (S, Γ_S, q) with conditional-law map $\Phi_S : S \rightarrow \mathcal{M}_0$, we obtain an induced semigroup $(P_t^S)_{t \geq 0}$ on observables over S , with generator L^S and resolvent $R^S(\lambda)$, by pullback along Φ_S .

Definition 26 (Dynamical neutral operator). *A dynamical neutral operator for Γ_Y and D_f is a quadruple*

$$\mathcal{N}_{\text{dyn}}(\Gamma_Y, D_f, (P_t^*)_{t \geq 0}) := (\Gamma_Y, \mathcal{R}(\Gamma_Y), \cong_{\text{IG}}, (P_t^*)_{t \geq 0}),$$

where $\mathcal{N}(\Gamma_Y, D_f)$ is the neutral operator from Definition 14, and $(P_t^*)_{t \geq 0}$ is a semigroup on $\mathcal{M}_0$ whose generator and resolvent are compatible with the canonical Fisher structure induced by D_f .

Theorem 6 (IG–spectral embedding). *Let*

$$(X, \Gamma_X, h), \quad (Z, \Gamma_Z, r) \in \mathcal{R}(\Gamma_Y)$$

be two realizations belonging to the same IG leaf $[S, \Gamma_S, q]_{\text{IG}}$ of the neutral operator $\mathcal{N}(\Gamma_Y, D_f)$. Let

$$\Phi_X : X \rightarrow \mathcal{M}_0, \quad \Phi_Z : Z \rightarrow \mathcal{M}_0$$

be their conditional-law maps into the canonical conditional-law manifold $(\mathcal{M}_0, g, \nabla, \nabla^*)$, and suppose that the induced semigroups $(P_t^X)_{t \geq 0}$ and $(P_t^Z)_{t \geq 0}$ on observables over X and Z are obtained by pullback from $(P_t^*)_{t \geq 0}$:

$$P_t^X f = f \circ (\Phi_X|_{X_0})^{-1} \circ P_t^* \circ \Phi_X, \quad P_t^Z g = g \circ (\Phi_Z|_{Z_0})^{-1} \circ P_t^* \circ \Phi_Z, \quad (2)$$

where $X_0 \subseteq X$ and $Z_0 \subseteq Z$ support the dynamics and the compositions are defined on the corresponding invariant subspaces.

Assume:

(S1) $\Phi_X|_{X_0}$ and $\Phi_Z|_{Z_0}$ are Fisher immersions of the same rank d with common image $\mathcal{M}_0$:

$$\Phi_X(X_0) = \Phi_Z(Z_0) = \mathcal{M}_0.$$

(S2) The pullback representation (2) holds for $\{P_t^X\}$ and $\{P_t^Z\}$ on the invariant subspaces generated by X_0 and Z_0 .

Then:

(a) There is a canonical dualistic isomorphism

$$\varphi := (\Phi_X|_{X_0})^{-1} \circ (\Phi_Z|_{Z_0}) : Z_0 \rightarrow X_0$$

between the supports, and the pullback $\varphi^\sharp$ on observables conjugates the generators:

$$\mathbb{L}^Z \circ \varphi^\sharp = \varphi^\sharp \circ \mathbb{L}^X$$

on the invariant subspaces linked by $\varphi^\sharp$.

(b) Their resolvents are likewise conjugate:

$$\mathbb{R}^Z(\lambda) \circ \varphi^\sharp = \varphi^\sharp \circ \mathbb{R}^X(\lambda)$$

for all λ in the common resolvent set of $\mathbb{L}^X$ and $\mathbb{L}^Z$.

(c) In particular, the spectral data agree:

$$\sigma(\mathbb{L}^X) = \sigma(\mathbb{L}^Z),$$

including algebraic multiplicities, on the invariant subspaces connected by $\varphi^\sharp$.

Remark 10 (Relation to the hypothesis form). *Theorem 6 is the conceptual form (“same IG leaf”). For later use it is sometimes convenient to work with an explicit hypothesis list; a sufficient set of such hypotheses is recorded below.*

Proposition 7 (IG-spectral embedding (hypothesis form)). *Let $\mathcal{N}_{\text{dyn}}(\Gamma_Y, D_f)$ be a dynamical neutral operator. Let (X, Γ_X, h) and (Z, Γ_Z, r) be realizations in $\mathcal{R}(\Gamma_Y)$ with conditional-law maps $\Phi_X : X \rightarrow \mathcal{M}_0$ and $\Phi_Z : Z \rightarrow \mathcal{M}_0$ and induced IG structures $(g_X, \nabla^X, (\nabla^X)^*)$ and $(g_Z, \nabla^Z, (\nabla^Z)^*)$.*

Assume there exist subsets $X_0 \subseteq X$ and $Z_0 \subseteq Z$ such that:

(I1) (IG-embedding hypotheses)

- $\Phi_X(X_0) = \Phi_Z(Z_0) = \mathcal{M}_0^{(0)}$ for some statistical submanifold $\mathcal{M}_0^{(0)} \subseteq \mathcal{M}_0$;
- the restrictions $\Phi_X|_{X_0} : X_0 \rightarrow \mathcal{M}_0^{(0)}$ and $\Phi_Z|_{Z_0} : Z_0 \rightarrow \mathcal{M}_0^{(0)}$ are Fisher immersions of the same rank.

(I2) (Dynamical compatibility) There exist Markov semigroups $(P_t^X)_{t \geq 0}$ and $(P_t^Z)_{t \geq 0}$ with generators L^X and L^Z and resolvents $R^X(\lambda)$ and $R^Z(\lambda)$ such that, on X_0 and Z_0 ,

$$\Phi_X \circ P_t^X = P_t^* \circ \Phi_X, \quad \Phi_Z \circ P_t^Z = P_t^* \circ \Phi_Z, \quad t \geq 0.$$

Then:

(a) There is a canonical dualistic isomorphism

$$\varphi := (\Phi_X|_{X_0})^{-1} \circ (\Phi_Z|_{Z_0}) : Z_0 \rightarrow X_0$$

(as in the IG-embedding theorem), which is a smooth coordinate changes onto its image and satisfies

$$g_Z = \varphi^* g_X, \quad \nabla^Z = \varphi^* \nabla^X, \quad (\nabla^Z)^* = \varphi^* (\nabla^X)^*$$

on Z_0 .

(b) The pullback map on observables

$$\varphi^\# : B_X \rightarrow B_Z, \quad (\varphi^\# f)(z) := f(\varphi(z)),$$

conjugates the semigroups:

$$P_t^Z \circ \varphi^\# = \varphi^\# \circ P_t^X, \quad t \geq 0,$$

and hence their generators and resolvents:

$$L^Z \circ \varphi^\# = \varphi^\# \circ L^X, \quad R^Z(\lambda) \circ \varphi^\# = \varphi^\# \circ R^X(\lambda)$$

for all λ in the common resolvent set.

(c) In particular, the spectra coincide on the invariant subspaces connected by $\varphi^\#$:

$$\sigma(L^Z) = \sigma(L^X),$$

including algebraic multiplicities, on those subspaces.

Remark 11 (Spectral stabilization inside an IG leaf). Once a pair of realizations lie in the same IG leaf and share a common dynamical neutral operator, Theorem 6 guarantees that their generators and resolvents are spectrally indistinguishable up to conjugacy on the relevant invariant subspaces. This is the abstract basis for defining spectrally faithful observation dimensions and for the stabilization principles used in the Rose hierarchy.

4.2 Abstract IG dynamical systems

Definition 27 (Information–geometric dynamical system). *Let $\mathcal{M}_0 \subset \mathcal{P}(Y)$ be a statistical manifold of probability measures on a measurable space Y , equipped with a dualistic structure (g, ∇, ∇^*) induced by a regular Csiszár–Amari f -divergence D_f .*

An information–geometric dynamical system (IGDS) on $(\mathcal{M}_0, g, \nabla, \nabla^)$ is a strongly continuous Markov semigroup $(P_t^*)_{t \geq 0}$ on $\mathcal{M}_0$ such that:*

(IG1) (Semigroup invariance) *For every $t \geq 0$, $P_t^* : \mathcal{M}_0 \rightarrow \mathcal{M}_0$ and $P_0^* = \text{id}_{\mathcal{M}_0}$, with $P_{t+s}^* = P_t^* \circ P_s^*$.*

(IG2) (Monotonicity of f -divergence) *For all $\mu, \nu \in \mathcal{M}_0$ and $t \geq 0$,*

$$D_f(P_t^* \mu \parallel P_t^* \nu) \leq D_f(\mu \parallel \nu).$$

(IG3) (Generator as a geometric vector field) *The infinitesimal generator L^* of (P_t^*) , defined on a suitable domain in $\mathcal{M}_0$, induces a (time-independent) smooth vector field*

$$X(\mu) := \left. \frac{d}{dt} P_t^* \mu \right|_{t=0} \in T_\mu \mathcal{M}_0,$$

and X is compatible with the dualistic structure in the sense that the covariant derivatives $\nabla_X X$ and $(\nabla^)_X X$ are well-defined tangent fields on $\mathcal{M}_0$.*

We write

$$(\mathcal{M}_0, g, \nabla, \nabla^*; (P_t^*)_{t \geq 0})$$

for such an information–geometric dynamical system, and we refer to L^ (or equivalently X) as its information–geometric generator.*

This definition packages:

- the *information* aspect via the divergence D_f and its monotonicity under (P_t^*) ;
- the *geometric* aspect via the dualistic structure (g, ∇, ∇^*) ;
- the *dynamical* aspect via the semigroup (P_t^*) and its generator L^* .

4.3 Empirical IG dynamical systems from the neutral operator

We now specialize this construction to the empirical setting of a fixed stepwise marginal Γ_Y and the neutral operator $\mathcal{N}(\Gamma_Y, D_f)$.

Definition 28 (Empirical IG dynamical system for Γ_Y). *Fix an empirical stepwise marginal $\Gamma_Y \in \mathcal{P}(Y \times Y)$ and a regular Csiszár–Amari f -divergence D_f on $\mathcal{P}(Y \times Y)$. Let $\mathcal{M}_0 \subset \mathcal{P}(Y)$ be a statistical manifold of conditional laws carrying the induced dualistic structure (g, ∇, ∇^*) .*

Assume that the one-step dynamics on Y extends to a Markov semigroup $(P_t^)_{t \geq 0}$ on $\mathcal{P}(Y)$ which:*

(EIG1) *leaves $\mathcal{M}_0$ invariant:*

$$P_t^*(\mathcal{M}_0) \subseteq \mathcal{M}_0 \quad \forall t \geq 0;$$

(EIG2) *is monotone for D_f on $\mathcal{M}_0$, in the sense that for all $\mu, \nu \in \mathcal{M}_0$ and $t \geq 0$,*

$$D_f(P_t^* \mu \parallel P_t^* \nu) \leq D_f(\mu \parallel \nu).$$

Then

$$(\mathcal{M}_0, g, \nabla, \nabla^*; (P_t^*)_{t \geq 0})$$

is an information–geometric dynamical system in the sense of Definition 27. We call this the empirical IG dynamical system associated with Γ_Y .

In the neutral-operator viewpoint, Γ_Y and D_f fix the IG kind of object, while the choice of (P_t^*) encodes the dynamical hypothesis for how conditional laws evolve in time.

4.4 Realizations as IG dynamical subbundles

Finally, we relate realizations in $\mathcal{R}(\Gamma_Y)$ to immersed IG dynamical subsystems of the empirical IGDS on $\mathcal{M}_0$.

Proposition 8 (Realizations as IG dynamical subbundles). *Let Γ_Y , D_f , $\mathcal{M}_0$ and (P_t^*) be as in Definition 28, and let*

$$(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$$

be a realization with conditional-law map $\Phi_S : S \rightarrow \mathcal{M}_0$ and induced IG structure $(g_S, \nabla^S, (\nabla^S)^)$. Suppose that:*

(R1) *the image $\Phi_S(S)$ is contained in $\mathcal{M}_0$ and supports the dynamics, in the sense that*

$$P_t^*(\Phi_S(S)) \subseteq \mathcal{M}_0 \quad \forall t \geq 0;$$

(R2) *the induced Fisher metric $g_S := \Phi_S^* g$ has constant rank d on a subset $S_0 \subseteq S$, so that $\Phi_S|_{S_0} : S_0 \rightarrow \mathcal{M}_0$ is a Fisher immersion of rank d ;*

(R3) *there exists a Markov semigroup $(P_t^S)_{t \geq 0}$ on S (induced by Γ_S) such that for all $t \geq 0$,*

$$\Phi_S \circ P_t^S = P_t^* \circ \Phi_S \quad \text{on } S_0.$$

Then:

- Φ_S identifies $(S_0, g_S, \nabla^S, (\nabla^S)^*)$ with a d -dimensional immersed submanifold of $(\mathcal{M}_0, g, \nabla, \nabla^*)$;
- the generator L^* of (P_t^*) restricts along Φ_S to a vector field X^S on S_0 , giving an induced IG dynamical system

$$(S_0, g_S, \nabla^S, (\nabla^S)^*; (P_t^S)_{t \geq 0})$$

that is information–geometrically equivalent to the restriction of $(\mathcal{M}_0, g, \nabla, \nabla^; (P_t^*))$ to $\Phi_S(S_0)$.*

In particular, each IG leaf in $\mathcal{R}(\Gamma_Y)$ selects, via its conditional-law manifold, a distinguished immersed IG dynamical subsystem of the empirical IG dynamical system on $\mathcal{M}_0$.

4.5 IG embeddings for observation families

We first formalize families of observation-based realizations and their information–geometric embeddings.

Definition 29 (Observation realization family). *Let $\Gamma_Y \in \mathcal{P}(Y \times Y)$ be a fixed empirical operator. An observation realization family for Γ_Y is a collection*

$$\{(Z^{(k)}, \Gamma_{Z^{(k)}}, r_k)\}_{k \in \mathcal{I}} \subset \mathcal{R}(\Gamma_Y),$$

indexed by a set $\mathcal{I} \subseteq \mathbb{N}$, such that for each $k \in \mathcal{I}$:

(O1) $Z^{(k)}$ is a measurable space;

(O2) $r_k : Y \rightarrow Z^{(k)}$ is a measurable observation map, typically of the form

$$r_k(y) = (h_1(y), \dots, h_k(y))$$

for scalar observation functions $h_i : Y \rightarrow \mathbb{R}$;

(O3) the stepwise marginal on $Z^{(k)}$ is induced by Γ_Y :

$$\Gamma_{Z^{(k)}} := (r_k \times r_k)_\# \Gamma_Y \in \mathcal{P}(Z^{(k)} \times Z^{(k)}).$$

Each triple $(Z^{(k)}, \Gamma_{Z^{(k)}}, r_k)$ is a representation in $\mathcal{R}(\Gamma_Y)$.

Assumption 1 (Reference realization). *There exists a reference realization*

$$(X, \Gamma_X, h_X) \in \mathcal{R}(\Gamma_Y)$$

(e.g. a latent dynamical model) with conditional-law map $\Phi_X : X \rightarrow \mathcal{P}(Y)$ and induced dualistic structure $(g_X, \nabla^X, (\nabla^X)^*)$ obtained by pulling back the Fisher–Rao dualistic structure on a statistical manifold $\mathcal{M}_0 \subset \mathcal{P}(Y)$.

Assumption 2 (IG compatibility of an observation index). *Let $k \in \mathcal{I}$. We say that k is IG-compatible with the reference realization if there exist:*

- subsets $X_0 \subseteq X$ and $Z_0^{(k)} \subseteq Z^{(k)}$ supporting the dynamics;
- a statistical manifold $\mathcal{M}_0 \subseteq \mathcal{P}(Y)$ containing the images of the conditional-law maps;
- conditional-law maps $\Phi_{Z^{(k)}} : Z^{(k)} \rightarrow \mathcal{M}_0$

such that:

$$(IC1) \quad \Phi_X(X_0) = \Phi_{Z^{(k)}}(Z_0^{(k)}) = \mathcal{M}_0;$$

(IC2) the restricted maps

$$\Phi_X|_{X_0} : X_0 \rightarrow \mathcal{M}_0, \quad \Phi_{Z^{(k)}}|_{Z_0^{(k)}} : Z_0^{(k)} \rightarrow \mathcal{M}_0$$

are Fisher immersions of the same rank d .

Theorem 7 (IG–multivariate observation embedding). *Let Assumption 1 hold and let $\{(Z^{(k)}, \Gamma_{Z^{(k)}}, r_k)\}_{k \in \mathcal{I}}$ be an observation realization family for Γ_Y . Suppose $k \in \mathcal{I}$ is IG-compatible in the sense of Assumption 2. Then:*

(a) *There is a canonical smooth map*

$$\varphi_k := (\Phi_X|_{X_0})^{-1} \circ (\Phi_{Z^{(k)}}|_{Z_0^{(k)}}) : Z_0^{(k)} \longrightarrow X_0$$

which is a smooth coordinate changes onto its image, with smooth inverse on each connected component supporting the dynamics.

(b) The map φ_k is a dualistic isomorphism between the induced IG structures:

$$g_{Z^{(k)}} = \varphi_k^* g_X, \quad \nabla^{Z^{(k)}} = \varphi_k^* \nabla^X, \quad (\nabla^{Z^{(k)}})^* = \varphi_k^* (\nabla^X)^*$$

on $Z_0^{(k)}$.

(c) The restricted stepwise marginals are related by pushforward:

$$\Gamma_{Z^{(k)}}|_{Z_0^{(k)} \times Z_0^{(k)}} = (\varphi_k^{-1} \times \varphi_k^{-1})_{\#} (\Gamma_X|_{X_0 \times X_0}).$$

In particular, $(Z^{(k)}, \Gamma_{Z^{(k)}}, r_k)$ lies in the same IG leaf as (X, Γ_X, h_X) on the supports $(Z_0^{(k)}, X_0)$.

Proof. This is a direct application of the IG-embedding theorem (Theorem 4) with $S = X$, $S' = Z^{(k)}$ and the conditional-law maps $\Phi_X, \Phi_{Z^{(k)}}$. Assumption 2 provides the hypotheses (image equality and Fisher immersions of the same rank), and the conclusions follow by specialization of Theorem 4. $\square$

Remark 12 (Effective observation dimension). *Under the hypotheses of Theorem 7, the common image $\mathcal{M}_0$ is a d -dimensional statistical manifold. Since $\Phi_X|_{X_0}$ and $\Phi_{Z^{(k)}}|_{Z_0^{(k)}}$ are Fisher immersions of rank d , they are local smooth coordinate changes onto $\mathcal{M}_0$, and hence*

$$\dim X_0 = \dim Z_0^{(k)} = \dim \mathcal{M}_0 = d.$$

Any additional coordinates in X or $Z^{(k)}$ along which the pullback Fisher metric is degenerate are Fisher-null directions and do not affect the IG leaf.

4.6 Spectrally faithful embeddings and spectral stabilization

We now add dynamics on top of the IG structure and define spectrally faithful observation dimensions inside a dynamical neutral operator.

Assumption 3 (Dynamical neutral operator). *Fix a regular Csiszár–Amari f -divergence D_f on $\mathcal{P}(Y \times Y)$. A dynamical neutral operator*

$$\mathcal{N}_{\text{dyn}}(\Gamma_Y, D_f)$$

consists of the neutral operator $\mathcal{N}(\Gamma_Y, D_f)$ together with, for each realization $(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$, a Markov semigroup $(P_t^S)_{t \geq 0}$ (or $(P_n^S)_{n \in \mathbb{N}}$) on a Banach space of observables over S , with generator L^S and resolvent $\mathsf{R}^S(\lambda)$, such that these are compatible with the pushforward structure along conditional-law maps and IG-equivalences.

Definition 30 (Spectrally faithful observation embedding dimension). *Let $\mathcal{N}_{\text{dyn}}(\Gamma_Y, D_f)$ be a dynamical neutral operator, and let (X, Γ_X, h_X) be a reference realization as in Assumption 1, with generator L^X and resolvent $\mathsf{R}^X(\lambda)$.*

Let $\{(Z^{(k)}, \Gamma_{Z^{(k)}}, r_k)\}_{k \in \mathcal{I}}$ be an observation realization family for Γ_Y , with generators $\mathsf{L}^{(k)}$ and resolvents $\mathsf{R}^{(k)}(\lambda)$ on appropriate Banach spaces of observables.

An index $k \in \mathcal{I}$ is called a Spectrally faithful observation embedding dimension (relative to X) if:

(SC1) k is IG-compatible in the sense of Assumption 2, yielding a dualistic isomorphism

$$\varphi_k := (\Phi_X|_{X_0})^{-1} \circ (\Phi_{Z^{(k)}}|_{Z_0^{(k)}}) : Z_0^{(k)} \rightarrow X_0$$

as in Theorem 7;

(SC2) the induced pullback map

$$\varphi_k^\sharp : \mathcal{B}(X_0) \rightarrow \mathcal{B}(Z_0^{(k)})$$

between the corresponding Banach spaces of observables intertwines generators and resolvents:

$$\mathbf{L}^{(k)} \circ \varphi_k^\sharp = \varphi_k^\sharp \circ \mathbf{L}^X, \quad \mathbf{R}^{(k)}(\lambda) \circ \varphi_k^\sharp = \varphi_k^\sharp \circ \mathbf{R}^X(\lambda)$$

for all λ in the common resolvent set.

In this case we say that $(Z^{(k)}, \Gamma_{Z^{(k)}}, r_k)$ is a Spectrally faithful observation embedding of the reference realization.

Definition 31 (Minimal Spectrally faithful observation dimension). Assume that there exists at least one Spectrally faithful observation dimension $k \in \mathcal{I}$ in the sense of Definition 30. The minimal Spectrally faithful observation embedding dimension (relative to X) is

$$k_* := \inf\{k \in \mathcal{I} : k \text{ is Spectrally faithful}\}.$$

If no such k exists, we say that the reference realization has no Spectrally faithful observation embedding within the chosen observation family.

Proposition 9 (Spectral stabilization across sufficiently rich observations). Let $\mathcal{N}_{\text{dyn}}(\Gamma_Y, D_f)$, (X, Γ_X, h_X) , and $\{(Z^{(k)}, \Gamma_{Z^{(k)}}, r_k)\}_{k \in \mathcal{I}}$ be as above. Suppose there exists $k_* \in \mathcal{I}$ such that every $k \in \mathcal{I}$ with $k \geq k_*$ is a Spectrally faithful observation embedding dimension.

Then for every $k \geq k_*$:

(a) The generators $\mathbf{L}^{(k)}$ and $\mathbf{L}^X$ are conjugate on the invariant subspaces connected by $\varphi_k^\sharp$:

$$\mathbf{L}^{(k)} \circ \varphi_k^\sharp = \varphi_k^\sharp \circ \mathbf{L}^X.$$

(b) Their resolvents are conjugate:

$$\mathbf{R}^{(k)}(\lambda) \circ \varphi_k^\sharp = \varphi_k^\sharp \circ \mathbf{R}^X(\lambda)$$

for all λ in the common resolvent set.

(c) Consequently, the spectral data stabilize:

$$\sigma(\mathbf{L}^{(k)}) = \sigma(\mathbf{L}^X),$$

including algebraic multiplicities, on the invariant subspaces linked by $\varphi_k^\sharp$.

Moreover, for any $k_1, k_2 \geq k_*$ there exists a dualistic isomorphism

$$\psi_{k_1, k_2} := \varphi_{k_1}^{-1} \circ \varphi_{k_2} : Z_0^{(k_2)} \rightarrow Z_0^{(k_1)}$$

such that the induced map $\psi_{k_1, k_2}^\sharp$ on observables conjugates the corresponding generators:

$$\mathbf{L}^{(k_2)} \circ \psi_{k_1, k_2}^\sharp = \psi_{k_1, k_2}^\sharp \circ \mathbf{L}^{(k_1)},$$

and hence

$$\sigma(\mathbf{L}^{(k_1)}) = \sigma(\mathbf{L}^{(k_2)})$$

on the connected invariant subspaces.

Proof. For each $k \geq k_*$, spectral faithfulness provides a dualistic isomorphism φ_k and a pullback map $\varphi_k^\#$ that conjugate $\mathbf{L}^{(k)}$ with $\mathbf{L}^X$ and $\mathbf{R}^{(k)}(\lambda)$ with $\mathbf{R}^X(\lambda)$. Conjugate operators have identical spectra (including algebraic multiplicities) on the invariant subspaces connected by the intertwining map, which yields (a)–(c).

For $k_1, k_2 \geq k_*$, define

$$\psi_{k_1, k_2} := \varphi_{k_1}^{-1} \circ \varphi_{k_2} : Z_0^{(k_2)} \rightarrow Z_0^{(k_1)}.$$

Let $\psi_{k_1, k_2}^\#$ be the induced map on observables. Using conjugacy of $\mathbf{L}^{(k_1)}$ and $\mathbf{L}^X$, and of $\mathbf{L}^{(k_2)}$ and $\mathbf{L}^X$, we obtain

$$\mathbf{L}^{(k_2)} \circ \psi_{k_1, k_2}^\# = \psi_{k_1, k_2}^\# \circ \mathbf{L}^{(k_1)},$$

so the spectra of $\mathbf{L}^{(k_1)}$ and $\mathbf{L}^{(k_2)}$ coincide on the corresponding invariant subspaces. $\square$

Remark 13 (Delay embeddings as a special observation family). *Classical delay embeddings arise by taking $Z^{(m)} = Y^m$ and $r_m(y_t) = (y_t, y_{t-1}, \dots, y_{t-m+1})$ for $m \geq 1$, yielding a particular observation realization family in the sense of Definition 29. In this case, any $m \geq m_*$ that is Spectrally faithful in the sense of Definition 30 is a Spectrally faithful delay embedding dimension, and Proposition 9 specializes to a spectral stabilization principle across sufficiently long delays.*

Diagrams

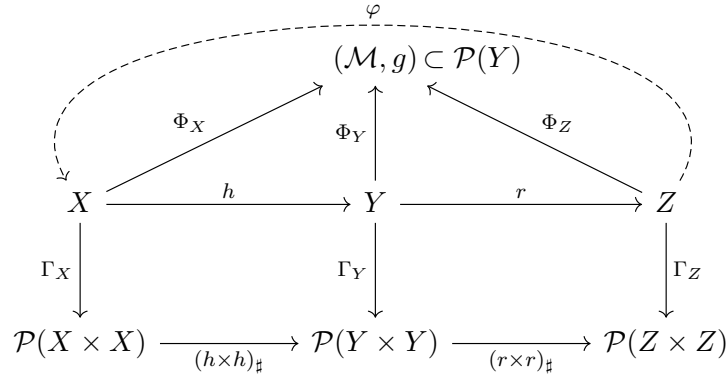


Figure 1: Realizations and stepwise marginals mapping by conditional-law maps Φ_X, Φ_Y, Φ_Z into a common information-geometric manifold $(\mathcal{M}, g) \subset \mathcal{P}(Y)$.

5 Optional strengthening: state-space faithfulness (Takens-type embeddings)

Roadmap. Everything above is decoder-free and pushforward-only. In this optional section we add state-space hypotheses (delay-embedding invertibility and/or sufficiency conditions) that promote boundary agreement into stronger relations between latent and representation realizations. The core development of this note is *boundary-first*: the boundary record $\Gamma_Y \in \mathcal{P}(Y \times Y)$, compatibility fibres, IG leaves, and completion as a closure/limit are defined without assuming an underlying state space, invertibility of representations, or a Markov semigroup *a priori*. Nevertheless, many implementations use delay coordinates or other rich representations where classical

embedding and conjugacy results provide an additional interpretation: certain realizations can be regarded as state-space reconstructions (up to smooth coordinate change) on the set supporting the dynamics.

5.1 Delays as path functionals vs state augmentation

A delay representation is formally a measurable functional of the observed record, i.e. a map on histories. When convenient, we may equivalently “Markovize” by augmenting the observed state to Y^m so that the delay representation becomes an ordinary state map and the canonical readout $\pi_0 : Y^m \rightarrow Y$ is available. This augmentation is optional and does not change the boundary record; it only changes how we *represent* candidates in the compatibility fibre.

Remark 14 (Delay representations need no learned decoder). *Delay-coordinate representations are a canonical special case of Definition 7. Take $Z = Y^m$ and define the (history) representation process*

$$Z_t := (Y_t, Y_{t-1}, \dots, Y_{t-m+1}) \in Y^m.$$

Then the readout map can be chosen canonically as the projection onto the present coordinate,

$$\pi_0 : Y^m \rightarrow Y, \quad \pi_0(y_0, \dots, y_{m-1}) = y_0.$$

For any stepwise marginal $\Gamma_Z \in \mathcal{P}(Y^m \times Y^m)$ on the delay space, the induced boundary stepwise marginal is

$$(\pi_0 \times \pi_0)_\# \Gamma_Z \in \mathcal{P}(Y \times Y).$$

In particular, the empirical delay edge record satisfies $(\pi_0 \times \pi_0)_\# \hat{\Gamma}_{Y^m}^{(T)} = \hat{\Gamma}_Y^{(T)}$ (up to the usual finite-sample boundary effects), so boundary comparison can be performed entirely in $\mathcal{P}(Y \times Y)$ without introducing any additional decoder beyond π_0 .

5.2 Topological (state-space) fidelity as an optional refinement

The following definition records a sufficient condition under which a representation can be interpreted as faithful at the level of the supporting set (e.g. an attractor). It is *not* used elsewhere: it may be skipped without affecting fibres, leaves, or completion.

Definition 32 (Topological fidelity from Y). *A representation $r : Y \rightarrow Z$ has topological fidelity of effective dimension d if there exists a conditional-law map*

$$\Phi_Z : Z \rightarrow \mathcal{M}_0 \subset \mathcal{P}(Y)$$

such that:

- (T1) Φ_Z is a Fisher immersion of rank d on a subset $Z_0 \subseteq Z$ supporting the dynamics;
- (T2) the pullback metric $g_Z = \Phi_Z^* g$ is nondegenerate on Z_0 , so that Z_0 inherits a d -dimensional manifold structure (up to negligible sets).

Delay embeddings as sufficient statistics (Čencov–Takens bridge)

The boundary-first framework never requires invertibility of a representation map. However, when a representation is *invertible on the support relevant to the model family*, Čencov-type information monotonicity upgrades from an inequality to an equality: the representation becomes a *sufficient statistic* for the chosen dynamical/statistical model family. This is the clean interface between Takens-style embedding hypotheses (state/representation level) and Čencov-style geometry preservation (information-geometric level).

Definition 33 (Sufficiency of a representation statistic). *Let $(\mathcal{P}_\theta)_{\theta \in \Theta}$ be a dominated family of probability laws on a data space Ω (for example, a family of path measures for a process, or a consistent family of finite-dimensional marginals encoding the same dynamics). Let $\rho : \Omega \rightarrow \Omega'$ be a measurable representation statistic and write $\mathcal{Q}_\theta := \rho_\# \mathcal{P}_\theta$ for the induced family on Ω' . We say that ρ is sufficient for $(\mathcal{P}_\theta)_{\theta \in \Theta}$ if there exists a Markov kernel $K : \Omega' \rightsquigarrow \Omega$, independent of θ , such that*

$$\mathcal{P}_\theta = K_\# \mathcal{Q}_\theta \quad \text{for all } \theta \in \Theta.$$

Equivalently, the conditional law of the original data given the statistic ρ , $\mathcal{P}_\theta(\cdot \mid r)$, is θ -independent.

Corollary 2 (Čencov/Takens upgrade: when the representation is lossless). *In the setting of Lemma 1:*

- (a) (Čencov/sufficiency regime) *If ρ is sufficient for the chosen statistical/dynamical model family in the sense of Definition 33, then the data-processing inequality becomes an equality on that family, so that working with $\Gamma_{Y,\rho}$ preserves both f -divergences and the induced Fisher geometry (up to the usual gauge/scale conventions).*
- (b) (Takens/invertibility regime) *If ρ is injective (or a diffeomorphic embedding) on the relevant support of the boundary record, then ρ may be treated as a change of coordinates on the supporting set. In this case, matching (or optimizing) candidates through the pushforward records on $Z \times Z$ is equivalent to matching them on $Y \times Y$, and one may transport realizations between Y -space and Z -space without introducing a separate decoder as a primitive assumption.*

Lemma 3 (Boundary Pythagorean decomposition (pushforward form)). *Let D be a divergence on $\mathcal{P}(Y \times Y)$ inducing a dualistic structure (g, ∇, ∇^*) on a regular statistical manifold containing the three boundary points*

$$p := (h \times h)_\# \Gamma_X, \quad q := \Gamma_Y, \quad r := (q_Z \times q_Z)_\# \Gamma_Z,$$

where (X, Γ_X, h) is a latent realization, and (Z, Γ_Z, q_Z) is a boundary-readable representation (for delay embeddings, one may take $q_Z = \pi_0$ on $Z = Y^m$). If q is the appropriate ∇ -projection of p onto a ∇^ -flat submanifold containing r (equivalently: the ∇ -geodesic from p to q is g -orthogonal at q to the ∇^* -geodesic from q to r), then*

$$D(p \parallel r) = D(p \parallel q) + D(q \parallel r).$$

In general, the deviation from orthogonality is measured by the standard dually-flat remainder term.

Proposition 10 (Čencov monotonicity and the equality case). *Fix a smooth statistical model $(\mathcal{P}_\theta)_{\theta \in \Theta}$ as above and let $I_\Omega(\theta)$ and $I_{\Omega'}(\theta)$ denote the Fisher information matrices (or Fisher metrics) for the families $(\mathcal{P}_\theta)$ and $(\mathcal{Q}_\theta)$, respectively. Then:*

- (a) (Monotonicity) $I_{\Omega'}(\theta) \preceq I_\Omega(\theta)$ for all $\theta \in \Theta$.

- (b) (Equality $\Leftrightarrow$ sufficiency) *Equality holds for all θ (in the sense of metric pullback/isometry on the model manifold) if and only if ρ is sufficient in the sense of Definition 33.*

In the equality case, the induced information-geometric structures on the two model manifolds are isomorphic (up to the usual gauge/scaling conventions), so that working with the statistic does not change the intrinsic Fisher geometry of the model family.

Lemma 4 (Čencov–Takens compatibility (as a regime, not a named theorem)). *Let (X, Γ_X, h) and (Z, Γ_Z, r) be realizations in $\mathcal{R}(\Gamma_Y)$ with conditional-law maps $\Phi_X : X \rightarrow \mathcal{M}_0$ and $\Phi_Z : Z \rightarrow \mathcal{M}_0$, where $\mathcal{M}_0$ carries the canonical Fisher metric g (Corollary 3). Assume:*

- (i) (Boundary consistency) *both realizations lie in the same compatibility fibre over Γ_Y (possibly in an ε -fibre), so that their boundary images agree with the boundary record in the sense fixed earlier.*
- (ii) (Takens-type invertibility on support) *there exist subsets $X_0 \subseteq X$ and $Z_0 \subseteq Z$ supporting the dynamics and a measurable bijection $\psi : Z_0 \rightarrow X_0$ such that, on Z_0 ,*

$$h = \pi_0 \circ r \quad \text{and} \quad \psi^{-1} \text{ implements the state reconstruction from } Z_0.$$

(For delay coordinates, $\pi_0 : Z \rightarrow Y$ is the canonical readout.)

- (iii) (Sufficiency for the induced statistical family) *the statistic induced by r (or by the corresponding delay/feature map on the data space) is sufficient for the chosen boundary/dynamical model family used to define the Fisher geometry on $\mathcal{M}_0$ (Definition 33), so that Proposition 10 applies.*

Then the induced information-geometric structures agree on the supported sets:

$$g_Z = \psi^* g_X, \quad \nabla^Z = \psi^* \nabla^X, \quad (\nabla^Z)^* = \psi^* (\nabla^X)^*,$$

and the two realizations lie in the same IG leaf of the neutral operator. In words: in the Takens invertibility regime, delay/representation embeddings act as sufficient statistics for the model family, so Fisher geometry is preserved rather than merely contracted.

5.3 What this buys you (and what it does not)

Under such hypotheses, some non-uniqueness inside the compatibility fibre may collapse: two realizations related by an embedding/conjugacy can be interpreted as alternative coordinate descriptions of the same underlying dynamics. This does *not* alter the boundary-first definitions: boundary fidelity is still assessed in $\mathcal{P}(Y \times Y)$, IG leaves are still defined by preserved information-geometric structure, and Markov completion remains a closure/limit statement.

A Dependency graph (who depends on what)

Roadmap. A quick “at a glance” index of dependencies among definitions and results. This list is meant as an “at a glance” dependency guide. References to kinds of objects are in parentheses.

- Definition 1 (B) depends on the notion of $\mathcal{P}(\cdot)$ and products of measurable spaces (see Appendix B).
- Definition 2 (B) depends on pushforward $(q \otimes q)_\#$ (Appendix B).

- Definition 3 (B) depends on Definition 2 and, for ε -fibres, on D_f (Definition 44, C).
- Definitions 4 and 44 (C) depend on dominated smooth families and f -divergences (Appendix B).
- Definitions 5 and Proposition 1 (C) depend on the dualistic/dually flat IG structure (Appendix B).
- Definition 23 (C over B) depends on choosing a boundary family $\mathcal{M}_Y$ and an IG notion of equivalence on that family (C).
- Definition 6 (B) depends on selecting a function space $\mathcal{X}$ and operator norms; it is the entry point for completion-as-closure.
- Proposition 2 (B) depends on Hille–Yosida/Trotter–Kato style hypotheses (standard functional analysis), invoked only when one wants a continuous-time completion.

B Background (moved to appendix)

Roadmap. Reference material and standard definitions used in the main text.

This section collects basic definitions and conventions used throughout.

B.1 Probability measures, pushforwards, and kernels

Definition 34 (Measurable space and probability measure). *A measurable space is a pair $(S, \mathcal{S})$ consisting of a set S and a σ -algebra $\mathcal{S} \subseteq 2^S$. A probability measure on $(S, \mathcal{S})$ is a map $\mu : \mathcal{S} \rightarrow [0, 1]$ that is countably additive and satisfies $\mu(S) = 1$. The set of all probability measures on $(S, \mathcal{S})$ is denoted $\mathcal{P}(S)$.*

Definition 35 (Pushforward measure). *Let $(S, \mathcal{S})$ and $(T, \mathcal{T})$ be measurable spaces, and let $f : S \rightarrow T$ be measurable. For $\mu \in \mathcal{P}(S)$ the pushforward of μ by f is the measure $f_{\#}\mu \in \mathcal{P}(T)$ defined by*

$$f_{\#}\mu(B) := \mu(f^{-1}(B)), \quad B \in \mathcal{T}.$$

If $\Gamma \in \mathcal{P}(S \times S)$ and $f : S \rightarrow T$, we write $(f \times f)_{\#}\Gamma$ for the pushforward on $T \times T$.

Definition 36 (Markov kernel). *Let $(S, \mathcal{S})$ and $(T, \mathcal{T})$ be measurable spaces. A Markov kernel from S to T is a map $K : S \times \mathcal{T} \rightarrow [0, 1]$ such that*

- *for each $A \in \mathcal{T}$, the map $s \mapsto K(s, A)$ is $\mathcal{S}$ -measurable;*
- *for each $s \in S$, the map $A \mapsto K(s, A)$ is a probability measure on $(T, \mathcal{T})$.*

Definition 37 (Stepwise marginal and disintegration). *Let $(S, \mathcal{S})$ be a measurable space and $\Gamma_S \in \mathcal{P}(S \times S)$ a probability measure. We call Γ_S a stepwise marginal on $S \times S$ if there exists a disintegration*

$$\Gamma_S(ds_0, ds_1) = \nu_S(ds_0) K_S(s_0, ds_1),$$

where $\nu_S \in \mathcal{P}(S)$ and K_S is a Markov kernel from S to S .

When $(S_t)_{t \in \mathbb{Z}}$ is a stochastic process with sample space S , the stepwise marginal of the process is

$$\Gamma_S := \mathbb{P}(S_t, S_{t+1}) \in \mathcal{P}(S \times S),$$

i.e. the joint law of two consecutive time points. In the time-series and graphical-model literatures this is often called the lag-1 joint distribution or an edge marginal on the edge $(t, t+1)$ of the time line; in the stationary case it coincides with the stationary pair measure of the Markov chain. We single it out because, in our framework, the stepwise marginal is the primitive transport object encoding one-step dynamics on S .

Lemma 5 (Stepwise marginals and conditional laws). *Let $(S, \mathcal{S})$ be a measurable space and $\Gamma_S \in \mathcal{P}(S \times S)$ a probability measure. Suppose Γ_S admits a disintegration*

$$\Gamma_S(ds_0, ds_1) = \nu_S(ds_0) K_S(s_0, ds_1),$$

with $\nu_S \in \mathcal{P}(S)$ and a Markov kernel K_S from S to S . Then:

- (a) *The pair (ν_S, K_S) is unique up to ν_S -null sets.*
- (b) *Conversely, any pair (ν_S, K_S) with $\nu_S \in \mathcal{P}(S)$ and a Markov kernel K_S defines a unique stepwise marginal $\Gamma_S \in \mathcal{P}(S \times S)$ via the above formula.*

In particular, there is a one-to-one correspondence (up to null sets) between stepwise marginals Γ_S and conditional-law maps $s \mapsto K_S(s, \cdot)$ together with a source law ν_S .

B.2 Statistical models and f -divergences

Definition 38 (Statistical model and statistical manifold). *Let $(\Omega, \mathcal{F})$ be a measurable space. A (parametric) statistical model on $(\Omega, \mathcal{F})$ is a family*

$$\mathcal{M} := \{\mu_\theta \in \mathcal{P}(\Omega) : \theta \in \Theta\},$$

where Θ is a smooth manifold and $\theta \mapsto \mu_\theta(A)$ is smooth for each $A \in \mathcal{F}$. We call Θ the parameter manifold, and the image $\mathcal{M} \subset \mathcal{P}(\Omega)$ (equipped with the induced smooth structure) the corresponding statistical manifold.

Definition 39 (Csiszár–Morimoto f -divergence). *Let $f : (0, \infty) \rightarrow \mathbb{R}$ be convex with $f(1) = 0$. For two probability measures $\mu, \nu \in \mathcal{P}(\Omega)$ such that $\mu \ll \nu$ and Radon–Nikodym derivative $\rho := \frac{d\mu}{d\nu}$, the f -divergence is*

$$D_f(\mu \| \nu) := \int f(\rho(\omega)) \nu(d\omega).$$

If $\mu \not\ll \nu$, we set $D_f(\mu \| \nu) := +\infty$.

Definition 40 (Csiszár–Amari α -divergences). *For $\alpha \in (-1, 1)$ and densities p, q on a common reference measure μ , the α -divergence is*

$$D_\alpha(p \| q) := \frac{4}{1 - \alpha^2} \left(1 - \int p^{\frac{1-\alpha}{2}} q^{\frac{1+\alpha}{2}} d\mu \right).$$

The limits $\alpha \rightarrow \pm 1$ yield forward and reverse Kullback–Leibler divergences. Each D_α is an f -divergence in the above sense.

In what follows any such divergence D_f on $\mathcal{P}(Y \times Y)$ serves both as a global discrepancy between stepwise marginals and as the generator of a local information geometry.

B.3 Fisher metric and dual affine connections

Let $\mathcal{M} = \{\mu_\theta : \theta \in \Theta\}$ be a smooth statistical model on $(\Omega, \mathcal{F})$, and let D_f be a smooth f -divergence.

Definition 41 (Fisher–Rao metric from D_f). *The Fisher–Rao metric g on Θ induced by D_f is the Riemannian metric whose components in a local coordinate chart $\theta = (\theta^1, \dots, \theta^d)$ are*

$$g_{ij}(\theta) := \frac{\partial^2}{\partial \epsilon_i \partial \epsilon_j} D_f(\mu_\theta \| \mu_{\theta+\epsilon}) \Big|_{\epsilon=0}, \quad 1 \leq i, j \leq d.$$

For regular f -divergences this coincides, up to a constant factor, with the classical Fisher information matrix whenever the μ_θ admit densities.

Definition 42 (Affine connection). *An affine connection ∇ on a smooth manifold Θ assigns to each pair of smooth vector fields X, Y a new vector field $\nabla_Y X$ such that ∇ is $\mathbb{R}$ -bilinear in (X, Y) , $C^\infty(\Theta)$ -linear in Y , and satisfies the Leibniz rule in X :*

$$\nabla_Y(fX) = Y(f)X + f\nabla_Y X$$

for all smooth functions f .

Definition 43 (Dual affine connections). *Let g be a Riemannian metric on Θ . Two affine connections ∇ and ∇^* are said to be dual with respect to g if for all smooth vector fields X, Y, Z on Θ ,*

$$Y\langle X, Z \rangle = \langle \nabla_Y X, Z \rangle + \langle X, \nabla_Y^* Z \rangle,$$

where $\langle \cdot, \cdot \rangle$ is the inner product induced by g .

Definition 44 (Dualistic structure). *A dualistic structure on a manifold Θ is a triple (g, ∇, ∇^*) consisting of a Riemannian metric g and two affine connections ∇, ∇^* that are dual with respect to g . For regular Csiszár–Amari α -divergences on a statistical model $\mathcal{M}$, the induced metric g and connections ∇, ∇^* are (up to overall scaling) independent of α .*

B.4 Monotone metrics and canonical Fisher geometry

We briefly record the uniqueness of the Fisher metric under Markov monotonicity, in the spirit of Čencov’s and Ay–Jost–Lê–Schwachhöfer’s results, and adapt it to the conditional-law manifold used in the neutral operator.

Definition 45 (Markov morphism between statistical models). *Let $(\Omega, \mathcal{F})$ and $(\Omega', \mathcal{F}')$ be measurable spaces, and let*

$$\mathcal{M} = \{\mu_\theta \in \mathcal{P}(\Omega) : \theta \in \Theta\}, \quad \mathcal{M}' = \{\mu'_\vartheta \in \mathcal{P}(\Omega') : \vartheta \in \Theta'\}$$

be statistical models with smooth parameter manifolds Θ, Θ' . A Markov morphism from $\mathcal{M}$ to $\mathcal{M}'$ is a pair (K, ψ) consisting of

- a Markov kernel $K : \Omega \times \mathcal{F}' \rightarrow [0, 1]$,
- a smooth map $\psi : \Theta' \rightarrow \Theta$,

such that for every $\vartheta \in \Theta'$,

$$\mu'_\vartheta = K_\# \mu_{\psi(\vartheta)},$$

where $K_\#$ denotes the pushforward of measures along K .

Definition 46 (Monotone Riemannian metric). *Let $\mathcal{M} = \{\mu_\theta\}$ be a statistical model with parameter manifold Θ . A Riemannian metric g on Θ is Markov monotone if for every Markov morphism $(K, \psi) : \mathcal{M} \rightarrow \mathcal{M}'$ between statistical models, with Riemannian metric g' on the parameter manifold Θ' , we have*

$$g'_\vartheta(v', v') \leq g_{\psi(\vartheta)}((d\psi_\vartheta v'), (d\psi_\vartheta v'))$$

for all $\vartheta \in \Theta'$ and all $v' \in T_\vartheta \Theta'$.

Theorem 8 (Čencov–Ay uniqueness of the Fisher metric). *Let $\mathcal{M} = \{\mu_\theta : \theta \in \Theta\}$ be a regular statistical model on a measurable space $(\Omega, \mathcal{F})$ dominated by a σ -finite reference measure, and let g be a smooth Riemannian metric on Θ that is Markov monotone in the sense above.*

Then g is proportional to the Fisher information metric g^F on Θ : there exists a constant $c > 0$ such that

$$g = c g^F.$$

Remark 15. *Theorem 8 is the continuous-measure counterpart of Čencov’s classical theorem for finite sample spaces: the Fisher metric is essentially the unique Riemannian metric that is monotone under Markov coarse-grainings and sufficient statistics. We use this result as a structural axiom for the neutral operator.*

We now adapt this to the conditional-law manifold associated with the stepwise marginal Γ_Y .

Definition 47 (Conditional-law manifold). *Let $\Gamma_Y \in \mathcal{P}(Y \times Y)$ be a stepwise marginal with disintegration*

$$\Gamma_Y(dy_0, dy_1) = \nu_Y(dy_0) K_Y(y_0, dy_1),$$

and let $\mathcal{R}(\Gamma_Y)$ be the realization space of Γ_Y . The conditional-law manifold $\mathcal{M}_0$ is any smooth statistical manifold

$$\mathcal{M}_0 \subset \mathcal{P}(Y)$$

containing the images of the conditional-law maps $\Phi_S : S \rightarrow \mathcal{P}(Y)$ for all realizations $(S, \Gamma_S, q) \in \mathcal{R}(\Gamma_Y)$, where

$$\Phi_S(s) := \mathbb{P}(Y_{t+1} \in \cdot \mid S_t = s).$$

Corollary 3 (Canonical IG structure on $\mathcal{M}_0$). *Let $\mathcal{M}_0 \subset \mathcal{P}(Y)$ be a conditional-law manifold as above. Suppose we require that any Riemannian metric g on $\mathcal{M}_0$ be Markov monotone with respect to Markov morphisms induced by measurable maps on Y , i.e. by Markov kernels of the form*

$$K_f(y, A) = \delta_{f(y)}(A), \quad f : Y \rightarrow Y',$$

and their pushforwards on conditional laws.

Then, up to an overall constant factor, there is a unique such metric g on $\mathcal{M}_0$, and g coincides with the Fisher–Rao metric induced by any regular Csiszár–Morimoto f -divergence on $\mathcal{P}(Y)$ restricted to $\mathcal{M}_0$.

Remark 16. *In particular, once Markov monotonicity is imposed, the choice of f -divergence on $\mathcal{P}(Y)$ does not change the information–geometric structure on $\mathcal{M}_0$ beyond an overall scale, in agreement with the α -neutrality theorem for Csiszár–Amari divergences.*

We finally record a compatibility statement that combines this canonical Fisher geometry with observation embeddings, including delay embeddings as a special case.

B.5 Manifolds, immersions, and foliations

Definition 48 (Smooth manifold). A (finite-dimensional) smooth manifold M of dimension d is a Hausdorff topological space equipped with an atlas of coordinate charts $\{(U_\alpha, \varphi_\alpha)\}$ such that each φ_α maps U_α homeomorphically onto an open subset of $\mathbb{R}^d$, and all transition maps $\varphi_\beta \circ \varphi_\alpha^{-1}$ are smooth wherever defined.

Definition 49 (Immersion). Let M and N be smooth manifolds, and let $F : M \rightarrow N$ be a smooth map. We say that F is an immersion at a point $x \in M$ if its differential

$$dF_x : T_x M \longrightarrow T_{F(x)} N$$

is injective. If F is an immersion at every point of M , then F is called an immersion.

Geometrically, an immersion has no “tangential collapse”: it may fail to be injective as a map $F : M \rightarrow N$, but it is locally an embedding at the level of tangent spaces.

Definition 50 (Vector bundle). Let M be a smooth manifold. A (real) vector bundle over M is a smooth surjective map $\pi : E \rightarrow M$ together with a vector space structure on each fibre $E_x := \pi^{-1}(\{x\})$ such that, for every $x \in M$, there is a neighbourhood $U \ni x$ and a smooth coordinate changes

$$\Phi : \pi^{-1}(U) \longrightarrow U \times \mathbb{R}^k$$

for some fixed k , with $\pi = \text{pr}_U \circ \Phi$, and such that each restriction $\Phi|_{E_x}$ is a linear isomorphism $E_x \cong \{x\} \times \mathbb{R}^k$. The integer k is the rank of the bundle. The tangent bundle $\pi : TM \rightarrow M$ is the canonical example.

Definition 51 (Subbundle). Let $\pi : E \rightarrow M$ be a vector bundle of rank k . A subbundle of E is a smooth submanifold $F \subset E$ such that:

- the restriction $\pi|_F : F \rightarrow M$ is a surjective submersion;
- for each $x \in M$, the fibre $F_x := F \cap E_x$ is a linear subspace of E_x of constant dimension r (independent of x).

In this case $\pi|_F : F \rightarrow M$ is itself a vector bundle of rank r , called a subbundle of E . For example, a smoothly varying choice of subspaces $H_x \subset T_x M$ defines a subbundle $H \subset TM$.

Definition 52 (Foliation). . Let M be a smooth manifold of dimension n . A foliation of M of codimension q (and dimension $p = n - q$) is a decomposition of M into disjoint connected subsets $\{L_\alpha\}_{\alpha \in A}$, called leaves, such that for every point $x \in M$ there exists a chart (U, φ) with

$$\varphi : U \rightarrow \mathbb{R}^p \times \mathbb{R}^q$$

a smooth coordinate changes onto an open set, and each nonempty connected component of $L_\alpha \cap U$ is mapped by φ onto a slice of the form

$$\mathbb{R}^p \times \{y_0\}, \quad y_0 \in \mathbb{R}^q.$$

In this case we say that the leaves $\{L_\alpha\}$ form a foliation of M .